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Anti-PD-1 antibodies and uses thereof

Abstract

This disclosure relates to isolated monoclonal antibodies or antigen-binding fragments thereof that specifically bind to PD-1 and block the interaction between PD-1 and PD-L1/PD-L2. The disclosure also relates to antibodies that are chimeric, humanized, bispecific, derivatized, single chain antibodies, portions of fusion proteins or bispecific antibodies. Nucleic acid molecules encoding the antibodies, hybridomas, and methods for expressing antibodies are also provided. Pharmaceutical compositions comprising the antibodies are also provided. This disclosure also provides use of these antibodies to enhance T-cell function and upregulate cell-mediated immune responses for the treatment and prevention of various diseases.

Inventors:	**He; Wei-Wu (Sugarland, TX), Bryson; Paul (Alhambra, CA), Li; Si (Alhambra, CA), Ma; Donghui (North Potomac, MD)**
Applicant:	**TCRCURE BIOPHARMA CORP.** (Chapel Hill, NC); **GUANGDONG TCRCURE BIOPHARMA TECHNOLOGY CO., LTD.** (Guangzhou, CN)
Family ID:	**72837646**
Assignee:	**TCRCURE BIOPHARMA CORP. (N/A, N/A); GUANGDONG TCRCURE BIOPHARMA TECHNOLOGY CO., LTD. (Guangzhou, CN)**
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Primary Examiner: Wu; Julie

Assistant Examiner: Heck; Bryan William

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

CLAIM OF PRIORITY (1) This application is a National Stage Application under 35 U.S.C. § 371 and claims the benefit of International Application No. PCT/US2020/028773, filed Apr. 17, 2020, which claims the benefit of U.S. Provisional Application No. 62/836,177, filed on Apr. 19, 2019. The entire contents of the foregoing are incorporated herein by reference.

TECHNICAL FIELD
(1) This disclosure relates to anti-PD-1 (Programmed Cell Death 1) antibodies, antigen-binding fragments, and the uses thereof.
BACKGROUND

(2) The protein Programmed Death 1 (PD-1 or PDCD1) is an inhibitory member of the CD28 family of receptors, which also includes CD28, CTLA-4, ICOS and BTLA. PD-1 is expressed on activated B cells, T cells, and myeloid cells (Agata et al., supra; Okazaki et al. (2002) Curr. Opin. Immunol. 14: 391779-82; Bennett et al. (2003) J Immunol 1 70:711-8). The PD-1 gene is a 55 kDa type I transmembrane protein that is part of the Ig gene superfamily (Agata et al. (1996) Int Immunol 8:765-72). PD-1 contains a membrane proximal immunoreceptor tyrosine inhibitory motif (ITIM) and a membrane distal tyrosine-based switch motif (ITSM) (Thomas, M. L. (1995) J Exp Med 181:1953-6; Vivier, E and Daeron, M (1997) Immunol Today 18:286-91). Although structurally similar to CTLA-4, PD-1 lacks the MYPPY motif that is critical for B7-1 and B7-2 binding.

(3) T-cells receive both positive and negative secondary co-stimulatory signals. The regulation of such positive and negative signals is critical to maximize the host's protective immune responses while maintaining immune tolerance and preventing autoimmunity. Negative secondary signals seem necessary for induction of T-cell tolerance, while positive signals promote T-cell activation. While the simple two-signal model provides a valid explanation for naive lymphocytes, a host's immune response is a dynamic process, and co-stimulatory signals can also be provided to antigen-exposed T-cells. PD-1 is an inhibitory member of the CD28 family expressed on activated B cells, T cells, and myeloid cells (Agata et al., supra; Okazaki et al. (2002) Curr Opin Immunol 14: 391779-82; Bennett et al. (2003) J Immunol 1 70:711-8) and provides negative signals.

(4) Two ligands for PD-1 have been identified, PD-L1 and PD-L2. They have been shown to downregulate T cell activation upon binding to PD-1 (Freeman et al. (2000) J Exp Med 192: 1027-34; Latchman et al. (2001) Nat Immunol 2:261-8; Carter et al. (2002) Eur J Immunol 32:634-43). Both PD-L1 and PD-L2 are B7 homologs that bind to PD-1, but do not bind to other CD28 family members. PD-L1 is abundant in a variety of human cancers (Dong et al. (2002) Nat. Med. 8:787-9). The interaction between PD-1 and PD-L1 results in a decrease in tumor infiltrating lymphocytes, a decrease in T-cell receptor-mediated proliferation, and immune evasion by the cancerous cells (Dong et al. (2003) J. Mal. Med. 81:281-7; Blank et al. (2005) Cancer Immunol. Immunother. 54:307-314; Konishi et al. (2004) Clin. Cancer Res. 10:5094-100). Immune suppression can be reversed by inhibiting the local interaction of PD-1 with PD-L1, and the effect is additive when the interaction of PD-1 with PD-L2 is blocked as well (Iwai et al. (2002) Proc. Nat'l. Acad. Sci. USA 99:12293-7; Brown et al. (2003) J. Immunol. 170: 1257-66).

(5) There exists a need in the field for antibodies or antigen-binding portions thereof, including chimeric and humanized antibodies, that neutralize/block PD-1 negative signal and stimulate an immune response, and that can have applications in the treatment of cancer and many other diseases.

SUMMARY

(6) The present disclosure provides monoclonal antibodies, or antigen-binding portions thereof that specifically bind to and neutralizes human PD-1. Monoclonal antibodies that are developed in mice can be immunogenic in humans. The present disclosure also provides humanized antibodies that were redesigned from the mouse antibody to reduce the immunogenicity in humans. These antibodies have improved efficacy and safety in humans.

(7) In one aspect, provided herein is an antibody or antigen-binding fragment thereof that binds to PD-1 (Programed Cell Death Protein 1) comprising a heavy chain variable region (VH) comprising complementarity determining regions (CDRs) 1, 2, and 3. In some embodiments, the VH CDR1 region comprises an amino acid sequence that is at least 80% identical to a selected VH CDR1 amino acid sequence, the VH CDR2 region comprises an amino acid sequence that is at least 80% identical to a selected VH CDR2 amino acid sequence, and the VH CDR3 region comprises an amino acid sequence that is at least 80% identical to a selected VH CDR3 amino acid sequence; and a light chain variable region (VL) comprising CDRs 1, 2, and 3. In some embodiments, the VL CDR1 region comprises an amino acid sequence that is at least 80% identical to a selected VL

CDR1 amino acid sequence, the VL CDR2 region comprises an amino acid sequence that is at least 80% identical to a selected VL CDR2 amino acid sequence, and the VL CDR3 region comprises an amino acid sequence that is at least 80% identical to a selected VL CDR3 amino acid sequence.

(8) In some embodiments, the selected VH CDRs 1, 2, and 3 amino acid sequences and the selected VL CDRs, 1, 2, and 3 amino acid sequences are one of the following: (1) the selected VH CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 8, 9, 10, respectively, and the selected VL CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 5, 6, 7, respectively; (2) the selected VH CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 22, 23, 24, respectively, and the selected VL CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 19, 20, 21, respectively; (3) the selected VH CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 74, 75, 76, respectively, and the selected VL CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 71, 72, 73, respectively; (4) the selected VH CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 34, 35, 36, respectively, and the selected VL CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 31, 32, 33, respectively; (5) the selected VH CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 54, 55, 56, respectively, and the selected VL CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 51, 52, 53, respectively; (6) the selected VH CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 64, 65, 66, respectively, and the selected VL CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 61, 62, 63, respectively; (7) the selected VH CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 44, 45, 46, respectively, and the selected VL CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 41, 42, 43, respectively.

(9) In some embodiments, the VH comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 8, 9, and 10, respectively, and the VL comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 5, 6, and 7, respectively.

(10) In some embodiments, the VH comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 22, 23, and 24, respectively, and the VL comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 19, 20, and 21, respectively.

(11) In some embodiments, the VH comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 74, 75, and 76, respectively, and the VL comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 71, 72, and 73, respectively.

(12) In some embodiments, the VH comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 34, 35, and 36, respectively, and the VL comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 31, 32, and 33, respectively.

(13) In some embodiments, the antibody or antigen-binding fragment specifically binds to human PD-1.

(14) In some embodiments, the antibody or antigen-binding fragment is a humanized antibody or antigen-binding fragment thereof.

(15) In some embodiments, the antibody or antigen-binding fragment is a single-chain variable fragment (scFv).

(16) In one aspect, provided herein is a nucleic acid comprising a polynucleotide encoding a polypeptide comprising: (1) an immunoglobulin heavy chain or a fragment thereof comprising a heavy chain variable region (VH) comprising complementarity determining regions (CDRs) 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 8, 9, and 10, respectively, and in some embodiments, the VH, when paired with a light chain variable region (VL) comprising the amino acid sequence set forth in SEQ ID NO: 2, 16, 28, 38, 48, 58 or 68 binds to PD-1; (2) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 5, 6, and 7, respectively, and in some embodiments, the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 4, 18, 30, 40, 50, 60, or 70 binds to PD-1; (3) an immunoglobulin heavy chain or a fragment thereof comprising a heavy chain variable region (VH) comprising CDRs 1, 2, and 3

comprising the amino acid sequences set forth in SEQ ID NOs: 22, 23, and 24, respectively, and in some embodiments, the VH, when paired with a light chain variable region (VL) comprising the amino acid sequence set forth in SEQ ID NO: 2, 16, 28, 38, 48, 58 or 68 binds to PD-1; (4) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 19, 20, and 21, respectively, and in some embodiments, the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 4, 18, 30, 40, 50, 60, or 70 binds to PD-1; (5) an immunoglobulin heavy chain or a fragment thereof comprising a heavy chain variable region (VH) comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 74, 75, and 76, respectively, and in some embodiments, the VH, when paired with a light chain variable region (VL) comprising the amino acid sequence set forth in SEQ ID NO: 2, 16, 28, 38, 48, 58 or 68 binds to PD-1; (6) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 71, 72, and 73, respectively, and in some embodiments, the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 4, 18, 30, 40, 50, 60, or 70 binds to PD-1; (7) an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 34, 35, and 36, respectively, and in some embodiments, the VH, when paired with a VL comprising the amino acid sequence set forth in SEQ ID NO: 2, 16, 28, 38, 48, 58 or 68 binds to PD-1; (8) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 31, 32, and 33, respectively, and in some embodiments, the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 4, 18, 30, 40, 50, 60, or 70 binds to PD-1; (9) an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 54, 55, and 56, respectively, and in some embodiments, the VH, when paired with a VL comprising the amino acid sequence set forth in SEQ ID NO: 2, 16, 28, 38, 48, 58 or 68 binds to PD-1; (10) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 51, 52, and 53, respectively, and in some embodiments, the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 4, 18, 30, 40, 50, 60, or 70 binds to PD-1; (11) an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 64, 65, and 66, respectively, and in some embodiments, the VH, when paired with a VL comprising the amino acid sequence set forth in SEQ ID NO: 2, 16, 28, 38, 48, 58 or 68 binds to PD-1; (12) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 61, 62, and 63, respectively, and in some embodiments, the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 4, 18, 30, 40, 50, 60, or 70 binds to PD-1; (13) an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 44, 45, and 46, respectively, and in some embodiments, the VH, when paired with a VL comprising the amino acid sequence set forth in SEQ ID NO: 2, 16, 28, 38, 48, 58 or 68 binds to PD-1; (14) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 41, 42, and 43, respectively, and in some embodiments, the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 4, 18, 30, 40, 50, 60, or 70 binds to PD-1.

(17) In some embodiments, the nucleic acid comprises a polynucleotide encoding a polypeptide comprising an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 8, 9, and 10, respectively.

(18) In some embodiments, the nucleic acid comprises a polynucleotide encoding a polypeptide

comprising an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 5, 6, and 7, respectively.
(19) In some embodiments, the nucleic acid comprises a polynucleotide encoding a polypeptide comprising an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 22, 23, and 24, respectively.
(20) In some embodiments, the nucleic acid comprises a polynucleotide encoding a polypeptide comprising an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 19, 20, and 21, respectively.
(21) In some embodiments, the nucleic acid comprises a polynucleotide encoding a polypeptide comprising an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 74, 75, and 76, respectively.
(22) In some embodiments, the nucleic acid comprises a polynucleotide encoding a polypeptide comprising an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 71, 72, and 73, respectively.
(23) In some embodiments, the nucleic acid comprises a polynucleotide encoding a polypeptide comprising an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 34, 35, and 36, respectively.
(24) In some embodiments, the nucleic acid comprises a polynucleotide encoding a polypeptide comprising an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 31, 32, and 33, respectively.
(25) In some embodiments, the VH when paired with a VL specifically binds to human PD-1, or the VL when paired with a VH specifically binds to human PD-1.
(26) In some embodiments, the immunoglobulin heavy chain or the fragment thereof is a humanized immunoglobulin heavy chain or a fragment thereof, and the immunoglobulin light chain or the fragment thereof is a humanized immunoglobulin light chain or a fragment thereof.
(27) In some embodiments, the nucleic acid encodes a single-chain variable fragment (scFv).
(28) In some embodiments, the nucleic acid is cDNA.
(29) In one aspect, provided herein is a vector comprising one or more of the nucleic acids as described herein.
(30) In one aspect, provided herein is a vector comprising two of the nucleic acids as described herein. In some embodiments, the vector encodes the VL region and the VH region that together bind to PD-1.
(31) In one aspect, provided herein is a pair of vectors. In some embodiments, each vector comprises one of the nucleic acids as described herein. In some embodiments, together the pair of vectors encodes the VL region and the VH region that together bind to PD-1.
(32) In one aspect, provided herein is a cell comprising the vector as described herein, or the pair of vectors as described herein.
(33) In some embodiments, the cell is a CHO cell.
(34) In one aspect, provided herein is a cell comprising one or more of the nucleic acids as described herein.
(35) In one aspect, provided herein a cell comprising two of the nucleic acids as described herein.
(36) In some embodiments, the two nucleic acids together encode the VL region and the VH region that together bind to PD-1.

(37) In one aspect, provided herein is a method of producing an antibody or an antigen-binding fragment thereof, the method comprising (a) culturing the cell as described herein under conditions sufficient for the cell to produce the antibody or the antigen-binding fragment; and (b) collecting the antibody or the antigen-binding fragment produced by the cell.
(38) In one aspect, provided herein is an antibody or antigen-binding fragment thereof that binds to PD-1 comprising a heavy chain variable region (VH) comprising an amino acid sequence that is at least 90% identical to a selected VH sequence, and a light chain variable region (VL) comprising an amino acid sequence that is at least 90% identical to a selected VL sequence. In some embodiments, the selected VH sequence is selected from SEQ ID NO: 4, 18, 30, 40, 50, 60, or 70, and the selected VL sequence is selected from SEQ ID NO: 2, 16, 28, 38, 48, 58 or 68.
(39) In some embodiments, the VH comprises the sequence of SEQ ID NO: 4 and the VL comprises the sequence of SEQ ID NO: 2.
(40) In some embodiments, the VH comprises the sequence of SEQ ID NO: 18 and the VL comprises the sequence of SEQ ID NO: 16.
(41) In some embodiments, the VH comprises the sequence of SEQ ID NO: 70 and the VL comprises the sequence of SEQ ID NO: 68.
(42) In some embodiments, the VH comprises the sequence of SEQ ID NO: 30 and the VL comprises the sequence of SEQ ID NO: 28.
(43) In some embodiments, the VH comprises the sequence of SEQ ID NO: 50 and the VL comprises the sequence of SEQ ID NO: 48.
(44) In some embodiments, the VH comprises the sequence of SEQ ID NO: 60 and the VL comprises the sequence of SEQ ID NO: 58.
(45) In some embodiments, the VH comprises the sequence of SEQ ID NO: 40 and the VL comprises the sequence of SEQ ID NO: 38.
(46) In some embodiments, the antibody or antigen-binding fragment specifically binds to human PD-1.
(47) In some embodiments, the antibody or antigen-binding fragment is a humanized antibody or antigen-binding fragment thereof.
(48) In some embodiments, the antibody or antigen-binding fragment is a single-chain variable fragment (scFv).
(49) In one aspect, provided herein is an antibody or antigen-binding fragment thereof comprising the VH CDRs 1, 2, 3, and the VL CDRs 1, 2, 3 of the antibody or antigen-binding fragment thereof as described herein.
(50) In one aspect, provided herein is an antibody-drug conjugate comprising the antibody or antigen-binding fragment thereof as described herein covalently bound to a therapeutic agent.
(51) In some embodiments, the therapeutic agent is a cytotoxic or cytostatic agent.
(52) In one aspect, provided herein is a method of treating a subject having cancer, the method comprising administering a therapeutically effective amount of a composition comprising the antibody or antigen-binding fragment thereof as described herein, or the antibody-drug conjugate as described herein, to the subject.
(53) In some embodiments, the subject has a solid tumor or a hematologic cancer (e.g., lymphoma).
(54) In some embodiments, the cancer is melanoma, non-small cell lung cancer, head and neck squamous cell carcinoma, relapsed or refractory classical Hodgkin lymphoma, squamous cell lung cancer, renal cell carcinoma, or cutaneous squamous cell carcinoma.
(55) In some embodiments, the cancer is urothelial carcinoma, merkel-cell carcinoma, or mesothelioma.
(56) In one aspect, provided herein is a method of decreasing the rate of tumor growth, the method comprising contacting an immune cell with an effective amount of a composition comprising an antibody or antigen-binding fragment thereof as described herein, or the antibody-drug conjugate as described herein, to the subject.

(57) In one aspect, provided herein is a method of killing a tumor cell, the method comprising contacting an immune cell with an effective amount of a composition comprising the antibody or antigen-binding fragment thereof as described herein, or the antibody-drug conjugate as described herein, to the subject.
(58) In one aspect, provided herein is a method of treating or reducing the risk of developing an infectious disease, the method comprising administering a therapeutically effective amount of a composition comprising the antibody or antigen-binding fragment thereof as described herein to the subject.
(59) In some embodiments, provided herein is the method that further comprises administering a vaccine to the subject.
(60) In one aspect, provided herein is a pharmaceutical composition comprising the antibody or antigen-binding fragment thereof as described herein, and a pharmaceutically acceptable carrier.
(61) In one aspect, provided herein is a pharmaceutical composition comprising the antibody drug conjugate as described herein, and a pharmaceutically acceptable carrier.
(62) In some embodiments, the antibody is an IgG1, IgG2, or IgG4 antibody. In some embodiments, the antibody is a human IgG1 antibody.
(63) In an aspect, an isolated monoclonal antibody, or antigen-binding portion thereof comprises: (a) a light chain variable domain CDR1 comprising SEQ ID NO:5; (b) a light chain variable domain CDR2 comprising SEQ ID NO:6; and (c) a light chain variable domain CDR3 comprising SEQ ID NO:7; (d) a heavy chain variable domain CDR1 comprising SEQ ID NO:8; (e) a heavy chain variable domain CDR2 comprising SEQ ID NO:9; (0 a heavy chain variable domain CDR3 comprising SEQ ID NO:10.
(64) In an aspect, the monoclonal antibody, or antigen-binding portion thereof blocks the interaction of PD-1 with both PD-L1 and PD-L2. Therefore the antibody, or antigen-binding portion can stimulate an anti-tumor immune response.
(65) In yet another aspect, the disclosure further provides a monoclonal antibody, or an antigen-binding portion thereof, which comprises: a light chain variable domain comprising SEQ ID NO:2 and a heavy chain variable domain comprising SEQ ID NO:4.
(66) The antibodies of the disclosure can be further engineered into formats suitable for human therapeutics by modifications that minimize immunogenicity. Suitable antibodies include, but are not limited to chimeric antibodies and humanized antibodies. The affinity, stability and specificity of the disclosed antibodies can also be further optimized by techniques known to one of skill in the art. Other formats can involve oligomerization, drug conjugation and fusion of the disclosed antibodies with other functional proteins.
(67) In yet another aspect, the humanized monoclonal antibody, or antigen-binding portion thereof, comprises a light chain variable domain amino acid sequence has at least 95% identity to SEQ ID NO:2 and a heavy chain variable domain amino acid sequence has at least 95% identity to SEQ ID NO:4.
(68) In yet another aspect, the disclosure further provides an isolated monoclonal antibody, or antigen-binding portion thereof comprising (a) a light chain variable domain CDR1 comprising SEQ ID NO: 5; (b) a light chain variable domain CDR2 comprising SEQ ID NO:6; (c) a light chain variable domain CDR3 comprising SEQ ID NO:7; and a heavy chain variable domain amino acid sequence with at least 95% identity to SEQ ID NO:4.
(69) In yet another aspect, the disclosure further provides an isolated monoclonal antibody, or antigen-binding portion thereof comprising (a) a heavy chain variable domain CDR1 comprising SEQ ID NO:8; (b) a heavy chain variable domain CDR2 comprising SEQ ID NO:9; (c) a heavy chain variable domain CDR3 comprising SEQ ID NO:10 and a light chain variable domain amino acid sequence with at least 95% identity to SEQ ID NO:2.
(70) The antibodies of the disclosure can be, for example, full-length antibodies, for example of an IgG1, IgG2, IgG3, or IgG4 isotype. Alternatively, the disclosed antibodies can be antibody

fragments, such as Fab, Fab′ and F(ab′)2 fragments, diabody, triabody, tetrabody, single-chain variable region fragment (scFv), disulfide-stabilized variable region fragment (dsFv), and half antibodies. Alternatively, the disclosed antibodies can be bispecific antibodies.
(71) In another aspect of the disclosure, an isolated monoclonal antibody or an antigen-binding portion thereof comprises: a light chain comprising SEQ ID NO:11 and a heavy chain comprising SEQ ID NO:12. In a preferred embodiment, the isolated monoclonal antibody comprises a light chain with the amino acid sequence of SEQ ID NO:13 and a heavy chain with the amino acid sequence of SEQ ID NO:14.
(72) In another aspect of the disclosure, a pharmaceutical composition comprising isolated monoclonal antibody, or antigen-binding portion thereof and a pharmaceutically acceptable carrier are also provided. Compositions comprising an immunoconjugate of the disclosure and a pharmaceutically acceptable carrier are also provided.
(73) In another aspect of the disclosure, an isolated nucleic acid molecule encoding the antibody, or antigen-binding portion thereof, and a host cell comprising a sequence of the nucleic acid molecule encoding the antibody, or antigen-binding portion thereof is also provided. In a preferred embodiment, the isolated nucleic acid molecule further comprises a sequence encoding a signal peptide. In a preferred embodiment, an isolated nucleic acid molecule encoding the antibody, or antigen-binding portion thereof, comprises the nucleic acid sequence of SEQ ID NO:1 and/or SEQ ID NO:3. In a preferred embodiment, the host cell is a hybridoma cell.
(74) The present disclosure further provides a method of stimulating immune responses using the anti-PD-1 antibodies of the disclosure. For example, in one embodiment, the disclosure provides a method for treating a subject in need thereof, comprising the step of administering to the subject an effective amount of the antibody or antigen-binding portion as described herein.
(75) In another aspect, the disclosure provides a method for treating cancer in a human comprising the step of administering to the human the antibody or antigen-binding portion as described herein in an amount effective to treat said cancer.
(76) In another aspect, the disclosure provides a method for treating infectious diseases in a human comprising the step of administering to the human the antibody or antigen-binding portion as described herein in an amount effective to treat said infectious diseases.
(77) In one aspect, the antibody or portion specifically binds to human PD-1 blocks the interaction between PD-1 and PD-L1.
(78) In one aspect, provided herein is a pharmaceutical composition comprising the antibody, or antigen-binding portion thereof as described herein and a pharmaceutically acceptable carrier.
(79) In one aspect, provided herein is a method of stimulating immune responses in a subject, comprising the step of administering to the subject the pharmaceutical composition as described herein in an amount effective to stimulate immune response in said subject.
(80) In one aspect, provided herein is a method of treating an infectious disease in a subject, comprising the step of administering to the subject the pharmaceutical composition as described herein in an amount effective to treat said infectious disease.
(81) In one aspect, provided herein is a method of treating cancer in a subject, comprising the step of administering to the subject the pharmaceutical composition as described herein in an amount effective to treat said cancer.
(82) In some embodiments, the monoclonal antibody, or antigen-binding portion thereof as described herein is a Fab fragment, an F(ab′)2 fragment, an Fv fragment, a single chain antibody, or a bispecific antibody.
(83) In some embodiments, the monoclonal antibody as described herein is a chimeric antibody or humanized antibody.
(84) In some embodiments, the monoclonal antibody as described herein is an immunoglobulin G (IgG), an IgM, an IgE, an IgA or an IgD molecule, or a derivative thereof In some embodiments, the monoclonal antibody described herein is an IgG1, IgG2, IgG3, or IgG4, or a derivative thereof.

(85) In some embodiments, the monoclonal antibody, or antigen-binding portion thereof as described herein blocks the interaction between PD-1 and PD-L2.
(86) In one aspect, provided herein is a monoclonal antibody comprising a light chain variable domain comprising the amino acid sequence of SEQ ID NO:2 and a heavy chain variable domain comprising the amino acid sequence of SEQ ID NO:4.
(87) In some embodiments, the monoclonal antibody as described herein is an immunoglobulin G (IgG), an IgM, an IgE, an IgA or an IgD molecule, or a derivative thereof. In some embodiments, the monoclonal antibody as described herein is an IgG1, IgG2, IgG3, or IgG4, or a derivative thereof.
(88) In some embodiments, the monoclonal antibody as described herein is a chimeric antibody.
(89) In some embodiments, the monoclonal antibody, or antigen-binding portion thereof as described herein comprises a light chain variable domain amino acid sequence having at least 95% identity to SEQ ID NO:2 and a heavy chain variable domain amino acid sequence having at least 95% identity to SEQ ID NO:4.
(90) In one aspect, provided herein is an isolated nucleic acid molecule encoding the antibody, or antigen-binding portion thereof as described herein.
(91) In one aspect, provided herein is a host cell comprising a sequence of the nucleic acid molecule as described herein.
(92) In one aspect, provided herein is an immunoconjugate comprising the antibody, or antigen-binding portion thereof as described herein, linked to a therapeutic agent.
(93) In one aspect, provided herein is a monoclonal antibody, or antigen-binding portion thereof, which comprises a light chain comprising SEQ ID NO:11 and a heavy chain comprising SEQ ID NO:12.
(94) In one aspect, the methods described herein can also be used for treating T cell dysfunctional disorders, including infection (e.g., acute and chronic) and tumor immunity.
(95) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Methods and materials are described herein for use in the present invention; other, suitable methods and materials known in the art can also be used. The materials, methods, and examples are illustrative only and not intended to be limiting. All publications, patent applications, patents, sequences, database entries, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control.
(96) Other features and advantages of the invention will be apparent from the following detailed description and figures, and from the claims.

Description

DESCRIPTION OF DRAWINGS
(1) Exemplary embodiments are illustrated in referenced figures. It is intended that the embodiments and figures disclosed herein are to be considered illustrative rather than restrictive.
(2) FIG. **1** shows the Western Blot analysis with the anti-PD-1 antibody (clone #11). HEK293T cells were transfected with the pCMV6-ENTRY control (Left lane) or pCMV6-ENTRY PDCD1 (RC210364, Right lane) cDNA for 48 hours and then were lysed. Equivalent amounts of cell lysates (5 μg per lane) were separated by SDS-PAGE and immunoblotted with anti-PD-1 antibodies (1:2000).
(3) FIG. **2**A shows the immunofluorescent staining of PDCD1 (RC210364)-stable-transfected HEK293T cells. PD-1 was labeled by mouse monoclonal anti-PD-1 antibody and the nucleus was labeled with Hoechst33342.
(4) FIG. **2**B shows the immunofluorescent staining of HEK293T cells as a negative control (1:100).

(5) FIG. **3**A shows immunocytochemistry staining of cells stably expressing PD-1 using the anti-PD-1 mouse monoclonal antibody (Left).
(6) FIG. **3**B shows immunocytochemistry staining of 293T cells as a negative control (1:900).
(7) FIG. **4** shows flow cytometric analysis of cells stably expressing PD-1 using anti-PD-1 antibody (F02) compared to a nonspecific negative control antibody (1:50).
(8) FIG. **5** shows flow cytometric analysis of cells stably expressing PD-L1 (RC213071) using anti-PD-1 antibody from hybridoma clone #11 (F02) or 0.3 ug/ml PD1-Fc fusion protein (TP700199) or both, and detected by anti-Fc (human) IgG-FITC (1:50). Binding of PD-L1 to PD-1 was completely blocked by anti-PD-1 antibody from hybridoma clone #11.
(9) FIG. **6** shows flow cytometric analysis of PD-L2(RC224141) transiently transfected HEK293T cells using anti-PD-1 antibody from hybridoma clone #11 (F02) or 1 μg/ml PD1-Fc fusion protein (TP700199) or both, and detected by anti-Fc (human) IgG-FITC (1:50). Binding of PD-L2 to PD-1 was completely blocked by anti-PD-1 antibody from hybridoma clone #11.
(10) FIG. **7**A shows a binding curve between F02 anti-PD-1 antibody and recombinant human PD-1 protein (rhPD-1). The binding affinity of F02 to rhPD-1 was determined by ELISA measurement. Kd value was determined as 0.14 nM.
(11) FIG. **7**B shows a binding curve between h11 anti-PD-1 antibody and rhPD-1. The binding affinity of h11 to rhPD-1 was determined by ELISA measurement. Kd value was determined as 0.29 nM.
(12) FIG. **7**C shows a binding curve between Ab2 anti-PD-1 antibody and rhPD-1.
(13) FIG. **7**D shows a binding curve between Ab4 anti-PD-1 antibody and rhPD-1.
(14) FIG. **7**E shows a binding curve between Ab5 anti-PD-1 antibody and rhPD-1.
(15) FIG. **7**F shows a binding curve between Ab7 anti-PD-1 antibody and rhPD-1. Kd value was determined as 0.68 nM.
(16) FIG. **7**G shows a binding curve between Ab8 anti-PD-1 antibody and rhPD-1. Kd value was determined as 0.32 nM.
(17) FIG. **8**A shows a binding curve between F02 anti-PD-1 antibody and human PD-1 protein (hPD-1) on cell surface. The binding affinity of F02 to hPD-1 on cell surface was determined by flow cytometry assessment. Kd value was determined as 0.07 nM.
(18) FIG. **8**B shows a binding curve between h11 anti-PD-1 antibody and human PD-1 protein (hPD-1) on cell surface. The binding affinity of h11 to hPD-1 on cell surface was determined by flow cytometry assessment. Kd value was determined as 0.55 nM.
(19) FIG. **8**C shows a binding curve between Ab2 anti-PD-1 antibody and human PD-1 protein (hPD-1) on cell surface. Kd value was determined as 3.24 nM.
(20) FIG. **8**D shows a binding curve between Ab4 anti-PD-1 antibody and human PD-1 protein (hPD-1) on cell surface. Kd value was determined as 41 nM.
(21) FIG. **8**E shows a binding curve between Ab5 anti-PD-1 antibody and human PD-1 protein (hPD-1) on cell surface. Kd value was determined as 3.69 nM.
(22) FIG. **8**F shows a binding curve between Ab7 anti-PD-1 antibody and human PD-1 protein (hPD-1) on cell surface. Kd value was determined as 2.3 nM.
(23) FIG. **8**G shows a binding curve between Ab8 anti-PD-1 antibody and human PD-1 protein (hPD-1) on cell surface. Kd value was determined as 0.65 nM.
(24) FIG. **9** lists sequences that are described in the present disclosure.

DETAILED DESCRIPTION

(25) The present disclosure relates to an isolated monoclonal antibody which binds to PD-1 and blocks interaction between PD-1 and PD-L1. In certain embodiments, the monoclonal antibody or antigen-binding portion thereof also blocks the interaction between PD-1 and PD-L2. In certain embodiments, the antibodies of the disclosure are derived from identified heavy and light chain germline sequences and/or comprise identified structural features such as CDR regions comprising identified amino acid sequences. This disclosure provides isolated antibodies, methods of making

such antibodies and antigen-binding portions thereof of the disclosure. This disclosure also relates to methods of using the antibodies, such as using the anti-PD-1 antibodies of the disclosure to stimulate immune responses, alone or in combination with other immunostimulatory antibodies. Accordingly, also provided are methods of using the anti-PD-1 antibodies of the disclosure for example, including but not limited to, treating cancer in a human.
(26) Therapeutic mouse antibodies sometimes can cause human anti-mouse antibody (HAMA) response. In this situation, therapeutic antibodies can lose their efficacy. The disclosure also provides humanized antibodies that have reduced HAMA response and prolonged therapeutic efficacy.
(27) In some embodiments, the humanized antibodies can be used in anticancer therapies. In some embodiments, the humanized antibodies can be used as a standalone therapeutic. In some embodiments, the humanized antibodies can be used as in combination with cell therapies. In some embodiments, the humanized antibodies described herein can be further refined into single-chain variable fragment (e.g., scFv).
(28) As used herein, the term “comprising” or “comprises” is used in reference to compositions, methods, and respective component(s) thereof, that are useful to an embodiment, yet open to the inclusion of unspecified elements, whether useful or not. It will be understood by those within the art that, in general, terms used herein are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.).
(29) Unless stated otherwise, the terms “a” and “an” and “the” and similar references used in the context of describing a particular embodiment of the application (especially in the context of claims) can be construed to cover both the singular and the plural. The recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (for example, “such as”) provided with respect to certain embodiments herein is intended merely to better illuminate the application and does not pose a limitation on the scope of the application otherwise claimed. The abbreviation, “e.g.” is derived from the Latin exempli gratia, and is used herein to indicate a non-limiting example. Thus, the abbreviation “e.g.” is synonymous with the term “for example.” No language in the specification should be construed as indicating any non-claimed element essential to the practice of the application.
(30) As used herein, the term “about” refers to a measurable value such as an amount, a time duration, and the like, and encompasses variations of ±20%, ±10%, ±5%, ±1%, ±0.5% or ±0.1% from the specified value.
(31) As used herein, the term “epitope” can include any protein determinant capable of specific binding to an immunoglobulin or T-cell receptor. Epitopic determinants usually consist of chemically active surface groupings of molecules such as amino acids or sugar side chains and usually have specific three-dimensional structural characteristics, as well as specific charge characteristics. An antibody is said to specifically bind an antigen when the equilibrium dissociation constant is <1 μM, preferably <100 nM and most preferably <10 nM.
(32) As used herein, the term “immune response” can refer to the action of, for example, lymphocytes, antigen presenting cells, phagocytic cells, granulocytes, and soluble macromolecules produced by the above cells or the liver (including antibodies, cytokines, and complement) that results in selective damage to, destruction of, or elimination from an organism of invading pathogens, cells or tissues infected with pathogens, cancerous cells, or, in cases of autoimmunity or pathological inflammation, normal organismal cells or tissues.

(33) As used herein, the term "antigen-specific T cell response" can refer to responses by a T cell that result from stimulation of the T cell with the antigen for which the T cell is specific. Non-limiting examples of responses by a T cell upon antigen-specific stimulation include, but are not limited to, proliferation and cytokine production (e.g., IL-2 production).
(34) As used herein, the term "antibody" refers to an intact immunoglobulin, or to a monoclonal or polyclonal antigen-binding fragment with the Fc (crystallizable fragment) region or FcRn binding fragment of the Fc region, referred to herein as the "Fc fragment" or "Fc domain". Antigen-binding fragments can be produced by recombinant DNA techniques or by enzymatic or chemical cleavage of intact antibodies. Antigen-binding fragments include, inter alia, Fab, Fab′, F(ab′)2, Fv, dAb, and complementarity determining region (CDR) fragments, single-chain antibodies (scFv), single domain antibodies, chimeric antibodies, diabodies and polypeptides that contain at least a portion of an immunoglobulin that is sufficient to confer specific antigen binding to the polypeptide. The Fc domain includes portions of two heavy chains contributing to two or three classes of the antibody. The Fc domain can be produced by recombinant DNA techniques or by enzymatic (e.g. papain cleavage) or via chemical cleavage of intact antibodies.
(35) As used herein, the term "antibody fragment" or "antigen-binding fragment" refers to a protein fragment that comprises only a portion of an intact antibody, generally including an antigen binding site of the intact antibody and thus retaining the ability to bind antigen. Examples of antibody fragments encompassed by the present definition include: (i) the Fab fragment, having VL, CL, VH and CH1 domains; (ii) the Fab′ fragment, which is a Fab fragment having one or more cysteine residues at the C-terminus of the CH1 domain; (iii) the Fd fragment having VH and CH1 domains; (iv) the Fd′ fragment having VH and CH1 domains and one or more cysteine residues at the C-terminus of the CH1 domain; (v) the Fv fragment having the VL and VH domains of a single arm of an antibody; (vi) the dAb fragment (Ward et al., Nature 341, 544-546 (1989)) which consists of a VH domain; (vii) isolated CDR regions; (viii) F(ab′)2 fragments, a bivalent fragment including two Fab′ fragments linked by a disulfide bridge at the hinge region; (ix) single chain antibody molecules (e.g., single chain Fv; scFv) (Bird et al., Science 242:423-426 (1988); and Huston et al., PNAS (USA) 85:5879-5883 (1988)); (x) "diabodies" with two antigen binding sites, comprising a heavy chain variable domain (VH) connected to a light chain variable domain (VL) in the same polypeptide chain (see, e.g., EP 404,097; WO 93/11161; and Hollinger et al., Proc. Natl. Acad. Sci. USA, 90:6444-6448 (1993)); (xi) "linear antibodies" comprising a pair of tandem Fd segments (VH-CH1-VH-CH1) which, together with complementary light chain polypeptides, form a pair of antigen binding regions (Zapata et al. Protein Eng. 8(10):1057-1062 (1995); and U.S. Pat. No. 5,641,870).
(36) As used herein, "single-chain variable fragment", "single-chain antibody variable fragments" or "scFv" antibodies refers to forms of antibodies comprising the variable regions of only the heavy (VH) and light (VL) chains, connected by a linker peptide. The scFvs are capable of being expressed as a single chain polypeptide. The scFvs retain the specificity of the intact antibody from which it is derived. The light and heavy chains can be in any order, for example, VH-linker-VL or VL-linker-VH, so long as the specificity of the scFv to the target antigen is retained.
(37) As used herein, an "isolated antibody" can refer to an antibody that is substantially free of other antibodies having different antigenic specificities (e.g., an isolated antibody that specifically binds a PD-1 protein can be substantially free of antibodies that specifically bind antigens other than PD-1 proteins). An isolated antibody that specifically binds a human PD-1 protein can, however, have cross-reactivity to other antigens, such as PD-1 proteins from other species. Moreover, an isolated antibody can be substantially free of other cellular material and/or chemicals.
(38) Anti-PD-1 antibody-producing cells, e.g., hybridomas, can be selected, cloned and further screened for desirable characteristics, including robust growth, high antibody production and desirable antibody characteristics. Hybridomas can be expanded in vivo in syngeneic animals, in animals that lack an immune system, e.g., nude mice, or in cell culture in vitro. Methods of

selecting, cloning and expanding hybridomas are well known to those of ordinary skill in the art.
(39) As used herein, the terms “monoclonal antibody” or “monoclonal antibody composition” can refer to a preparation of antibody molecules of single molecular composition. A monoclonal antibody composition displays a single binding specificity and affinity for a particular epitope.
(40) As used herein, the term “recombinant human antibody”, can refer to all human antibodies that are prepared, expressed, created or isolated by recombinant means, such as (a) antibodies isolated from an animal (e.g., a mouse) that is transgenic or transchromosomal for human immunoglobulin genes or a hybridoma prepared therefrom (described below), (b) antibodies isolated from a host cell transformed to express the human antibody, e.g., from a transfectoma, (c) antibodies isolated from a recombinant, combinatorial human antibody library, and (d) antibodies prepared, expressed, created or isolated by any other means that involve splicing of human immunoglobulin gene sequences to other DNA sequences. Such recombinant human antibodies have variable regions in which the framework and CDR regions are derived from human germline immunoglobulin sequences. In certain embodiments, however, such recombinant human antibodies can be subjected to in vitro mutagenesis (or, when an animal transgenic for human Ig sequences is used, in vivo somatic mutagenesis) and thus the amino acid sequences of the VH and VL regions of the recombinant antibodies are sequences that, while derived from and related to human germline VH and VL sequences, may not naturally exist within the human antibody germline repertoire in vivo.
(41) As used herein, the term “isotype” can refer to the antibody class (e.g., IgM or IgG1) that is encoded by the heavy chain constant region genes.
(42) A “humanized antibody” has a sequence that differs from the sequence of an antibody derived from a non-human species by one or more amino acid substitutions, deletions, and/or additions, such that the humanized antibody is less likely to induce an anti-antibody immune response, and/or induces a less severe anti-antibody immune response, as compared to the non-human species antibody, when it is administered to a human subject. In some embodiments, certain amino acids in the framework and constant domains of the heavy and/or light chains of the non-human species antibody are mutated to produce the humanized antibody. In some embodiments, the constant domain(s) from a human antibody are fused to the variable domain(s) of a non-human species. In some embodiments, one or more amino acid residues in one or more CDR sequences of a non-human antibody are changed to reduce the likely immunogenicity of the non-human antibody when it is administered to a human subject, wherein the changed amino acid residues either are not critical for immunospecific binding of the antibody to its antigen, or the changes to the amino acid sequence that are made are conservative changes, such that the binding of the humanized antibody to the antigen is not significantly worse than the binding of the non-human antibody to the antigen. Examples of how to make humanized antibodies can be found in U.S. Pat. Nos. 6,054,297, 5,886,152 and 5,877,293, which are incorporated herein by reference in the entirety.
(43) As used herein, the term “chimeric antibody” can refer to antibodies in which the variable region sequences can be derived from one species and the constant region sequences can be derived from another species, such as an antibody in which the variable region sequences can be derived from a mouse antibody and the constant region sequences can be derived from a human antibody.
(44) As used herein, an antibody that “specifically binds human PD-1” can refer to an antibody that binds to a human PD-1 protein (and possibly a PD-1 protein from one or more non-human species) but does not substantially bind to non-PD-1 proteins. In some embodiments, the antibody binds to a human PD-1 protein with “high affinity,” e.g., with a EC50 of 1×10.sup.−7 M or less, more preferably 5×10.sup.−8 M or less, more preferably 3×10.sup.−8 M or less, more preferably 1×10.sup.−8 M or less, more preferably 5×10.sup.−9M or less or even more preferably 1×10.sup.−9M or less. The phrases “an antibody recognizing an antigen” and “an antibody specific for an antigen” are used interchangeably herein with the term “an antibody which binds specifically to an antigen.”
(45) As used herein, the term “does not substantially bind” to a protein or cells, can mean that it

cannot bind or does not bind with a high affinity to the protein or cells, i.e., binds to the protein or cells with an EC50 of 2×10^{-6} M or more, more preferably 1×10^{-5}M or more, more preferably 1×10^{-4}M or more, more preferably 1×10^{-3} M or more, even more preferably 1×10^{-2} M or more.
(46) As used herein, the term “high affinity” for an IgG antibody can refer to an antibody having an Kd of 1×10^{-6}M or less, more preferably 1×10^{-7} M or less, even more preferably 1×10^{-8}M or less, even more preferably 1×10^{-9}M or less, even more preferably 1×10^{-10} M or less for a target antigen. However, “high affinity” binding can vary for other antibody isotypes.
(47) As used herein, the term “pharmaceutical formulation” refers to a preparation which is in such form as to permit the biological activity of an active ingredient contained therein to be effective, and which contains no additional components which are unacceptably toxic to a subject to which the formulation would be administered.
(48) As used herein, the term “inhibit” refers to any decrease in, for example a particular action, function, or interaction. For example, a biological function, such as the function of a protein and/or binding of one protein to another, is inhibited if it is decreased as compared to a reference state, such as a control like a wild-type state or a state in the absence of an applied agent. For example, the binding of a PD-1 protein to one or more of its ligands, such as PD-L1 and/or PD-L2, and/or resulting PD-1 signaling and immune effects is inhibited or deficient if the binding, signaling, and other immune effects are decreased due to contact with an agent, such as an anti-PD-1 antibody, in comparison to when the PD-1 protein is not contacted with the agent. Such inhibition or deficiency can be induced, such as by application of agent at a particular time and/or place, or can be constitutive, such as by continual administration. Such inhibition or deficiency can also be partial or complete (e.g., essentially no measurable activity in comparison to a reference state, such as a control like a wild-type state). Essentially complete inhibition or deficiency is referred to as blocked.
(49) As used herein, the term “subject” can refer to any human or non-human animal. The subject can be male or female and can be any suitable age, including infant, juvenile, adolescent, adult, and geriatric subjects. The term “nonhuman animal” includes all vertebrates, e.g., mammals and non-mammals, such as nonhuman primates, sheep, dogs, cats, cows, horses, chickens, rabbits, mice, rats, amphibians, and reptiles, although mammals are preferred, such as non-human primates, sheep, dogs, cats, cows and horses.
(50) PD-1 and Immune System
(51) The immune system can differentiate between normal cells in the body and those it sees as “foreign,” which allows the immune system to attack the foreign cells while leaving the normal cells alone. This mechanism sometimes involves proteins called immune checkpoints. Immune checkpoints are molecules in the immune system that either turn up a signal (co-stimulatory molecules) or turn down a signal.
(52) Checkpoint inhibitors can prevent the immune system from attacking normal tissue and thereby preventing autoimmune diseases. Many tumor cells also express checkpoint inhibitors. These tumor cells escape immune surveillance by co-opting certain immune-checkpoint pathways, particularly in T cells that are specific for tumor antigens (Creelan, Benjamin C. “Update on immune checkpoint inhibitors in lung cancer.” Cancer Control 21.1 (2014): 80-89). Because many immune checkpoints are initiated by ligand-receptor interactions, they can be readily blocked by antibodies against the ligands and/or their receptors.
(53) PD-1 (Programmed Cell Death 1; PDCD1; or Programmed Death 1) is an immune checkpoint and guards against autoimmunity through a dual mechanism of promoting apoptosis (programmed cell death) in antigen-specific T-cells in lymph nodes while simultaneously reducing apoptosis in regulatory T cells (anti-inflammatory, suppressive T cells).
(54) PD-1 is mainly expressed on the surfaces of T cells and primary B cells; two ligands of PD-1

(PD-L1 and PD-L2) are widely expressed in antigen-presenting cells (APCs). The interaction of PD-1 with its ligands plays an important role in the negative regulation of the immune response. Inhibition the binding between PD-1 and its ligand can make the tumor cells exposed to the killing effect of the immune system, and thus can reach the effect of killing tumor tissues and treating cancers.

(55) PD-L1 (CD274) is expressed on the neoplastic cells of many different cancers. By binding to PD-1 on T-cells leading to its inhibition, PD-L1 expression is a major mechanism by which tumor cells can evade immune attack. PD-L1 over-expression may conceptually be due to 2 mechanisms, intrinsic and adaptive. Intrinsic expression of PD-L1 on cancer cells is related to cellular/genetic aberrations in these neoplastic cells. Activation of cellular signaling including the AKT and STAT pathways results in increased PD-L1 expression. In primary mediastinal B-cell lymphomas, gene fusion of the MHC class II transactivator (CIITA) with PD-L1 or PD-L2 occurs, resulting in overexpression of these proteins. Amplification of chromosome 9p23-24, where PD-L1 and PD-L2 are located, leads to increased expression of both proteins in classical Hodgkin lymphoma. Adaptive mechanisms are related to induction of PD-L1 expression in the tumor microenvironment. PD-L1 can be induced on neoplastic cells in response to interferon γ. In microsatellite instability colon cancer, PD-L1 is mainly expressed on myeloid cells in the tumors, which then suppress cytotoxic T-cell function.

(56) The use of PD-1 blockade to enhance anti-tumor immunity originated from observations in chronic infection models, where preventing PD-1 interactions reversed T-cell exhaustion. Similarly, blockade of PD-1 prevents T-cell PD-1/tumor cell PD-L1 or T-cell PD-1/tumor cell PD-L2 interaction, leading to restoration of T-cell mediated anti-tumor immunity.

(57) A detailed description of PD-1, and the use of anti-PD-1 antibodies to treat cancers are described, e.g., in Topalian, Suzanne L., et al. “Safety, activity, and immune correlates of anti-PD-1 antibody in cancer.” New England Journal of Medicine 366.26 (2012): 2443-2454; Hirano, Fumiya, et al. “Blockade of B7-H1 and PD-1 by monoclonal antibodies potentiates cancer therapeutic immunity.” Cancer research 65.3 (2005): 1089-1096; Raedler, Lisa A. “Keytruda (pembrolizumab): first PD-1 inhibitor approved for previously treated unresectable or metastatic melanoma.” American health & drug benefits 8. Spec Feature (2015): 96; Kwok, Gerry, et al. “Pembrolizumab (Keytruda).” (2016): 2777-2789; US 20170247454; U.S. Pat. Nos. 9,834,606 B; and 8,728,474; each of which is incorporated by reference in its entirety.

(58) The present disclosure provides anti-PD-1 antibodies, antigen-binding fragments thereof, and methods of using these anti-PD-1 antibodies and antigen-binding fragments to inhibit tumor growth and to treat cancers.

(59) Anti-PD-1 Antibodies and Antigen-Binding Fragments

(60) The disclosure provides antibodies and antigen-binding fragments thereof that specifically bind to PD-1. The antibodies and antigen-binding fragments described herein are capable of binding to PD-1. In some embodiments, these antibodies can block PD-1 signaling pathway thus increase immune response. In some embodiments, these antibodies can initiate complement-dependent cytotoxicity (CMC) or antibody-dependent cellular cytotoxicity (ADCC).

(61) The disclosure provides e.g., mouse anti-PD-1 antibodies (e.g., F02), and the chimeric antibodies thereof, and the humanized antibodies thereof (e.g., H11, Ab8, Ab2, Ab5, Ab7, Ab4).

(62) The CDR sequences for F02, and F02 derived antibodies or antigen-binding fragments thereof include VH CDR1, VH CDR2, and VH CDR3 comprising or consisting of SEQ ID NOs: 8, 9, 10 respectively, and VL CDR1, VL CDR2, and VL CDR3 comprising or consisting of SEQ ID NOs: 5, 6, 7, respectively.

(63) The CDR sequences for H11, and H11 derived antibodies or antigen-binding fragments thereof include VH CDR1, VH CDR2, and VH CDR3 comprising or consisting of SEQ ID NOs: 22, 23, 24, respectively, and VL CDR1, VL CDR2, and VL CDR3 comprising or consisting of SEQ ID NOs: 19, 20, 21, respectively.

(64) The CDR sequences for Ab8, and Ab8 derived antibodies or antigen-binding fragments thereof include VH CDR1, VH CDR2, and VH CDR3 comprising or consisting of SEQ ID NOs: 74, 75, 76, respectively, and VL CDR1, VL CDR2, and VL CDR3 comprising or consisting of SEQ ID NOs: 71, 72, 73, respectively.
(65) The CDR sequences for Ab2, and Ab2 derived antibodies or antigen-binding fragments thereof include VH CDR1, VH CDR2, and VH CDR3 comprising or consisting of SEQ ID NOs: 34, 35, 36, respectively, and VL CDR1, VL CDR2, and VL CDR3 comprising or consisting of SEQ ID NOs: 31, 32, 33, respectively.
(66) The CDR sequences for Ab5, and Ab5 derived antibodies or antigen-binding fragments thereof include VH CDR1, VH CDR2, and VH CDR3 comprising or consisting of SEQ ID NOs: 54, 55, 56, respectively, and VL CDR1, VL CDR2, and VL CDR3 comprising or consisting of SEQ ID NOs: 51, 52, 53, respectively.
(67) The CDR sequences for Ab7, and Ab7 derived antibodies or antigen-binding fragments thereof include VH CDR1, VH CDR2, and VH CDR3 comprising or consisting of SEQ ID NOs: 64, 65, 66, respectively, and VL CDR1, VL CDR2, and VL CDR3 comprising or consisting of SEQ ID NOs: 61, 62, 63, respectively.
(68) The CDR sequences for Ab8, and Ab8 derived antibodies or antigen-binding fragments thereof include VH CDR1, VH CDR2, and VH CDR3 comprising or consisting of SEQ ID NOs: 44, 45, 46, respectively, and VL CDR1, VL CDR2, and VL CDR3 comprising or consisting of SEQ ID NOs: 41, 42, 43, respectively.
(69) The amino acid sequence for the heavy chain variable region of F02 antibody is set forth in SEQ ID NO: 4. The amino acid sequence for the light chain variable region of F02 antibody is set forth in SEQ ID NO: 2.
(70) The amino acid sequence for the heavy chain variable region of H11 antibody is set forth in SEQ ID NO: 18. The amino acid sequence for the light chain variable region of H11 antibody is set forth in SEQ ID NO: 16.
(71) The amino acid sequence for the heavy chain variable region of Ab8 antibody is set forth in SEQ ID NO: 70. The amino acid sequence for the light chain variable region of Ab8 antibody is set forth in SEQ ID NO: 68.
(72) The amino acid sequence for the heavy chain variable region of Ab2 antibody is set forth in SEQ ID NO: 30. The amino acid sequence for the light chain variable region of Ab2 antibody is set forth in SEQ ID NO: 28.
(73) The amino acid sequence for the heavy chain variable region of Ab5 antibody is set forth in SEQ ID NO: 50. The amino acid sequence for the light chain variable region of Ab5 antibody is set forth in SEQ ID NO: 48.
(74) The amino acid sequence for the heavy chain variable region of Ab7 antibody is set forth in SEQ ID NO: 60. The amino acid sequence for the light chain variable region of Ab7 antibody is set forth in SEQ ID NO: 58.
(75) The amino acid sequence for the heavy chain variable region of Ab4 antibody is set forth in SEQ ID NO: 40. The amino acid sequence for the light chain variable region of Ab4 antibody is set forth in SEQ ID NO: 38.
(76) The amino acid sequences for heavy chain variable regions and light variable regions of the humanized antibodies are also provided. As there are different ways to humanize a mouse antibody (e.g., a sequence can be modified with different amino acid substitutions), the heavy chain and the light chain of an antibody can have more than one version of humanized sequences. In some embodiments, the humanized heavy chain variable region is at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identical to SEQ ID NO: 4, 18, 70, 30, 50, 60, or 40. In some embodiments, the humanized light chain variable region is at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identical to SEQ ID NO: 2, 16, 68, 28, 48, 58, or 38. The heavy chain variable region sequence can be paired with the light chain variable region

sequence, and together they bind to PD-1.

(77) Humanization percentage means the percentage identity of the heavy chain or light chain variable region sequence as compared to human antibody sequences in International Immunogenetics Information System (IMGT) database. The top hit means that the heavy chain or light chain variable region sequence is closer to a particular species than to other species. For example, top hit to human means that the sequence is closer to human than to other species. Top hit to human and *Macaca fascicularis* means that the sequence has the same percentage identity to the human sequence and the *Macaca fascicularis* sequence, and these percentages identities are highest as compared to the sequences of other species. In some embodiments, humanization percentage is greater than 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, or 95%. A detailed description regarding how to determine humanization percentage and how to determine top hits is known in the art, and is described, e.g., in Jones, et al. "The INNs and outs of antibody nonproprietary names." MAbs. Vol. 8. No. 1. Taylor & Francis, 2016, which is incorporated herein by reference in its entirety. A high humanization percentage often has various advantages, e.g., more safe and more effective in humans, more likely to be tolerated by a human subject, and/or less likely to have side effects.

(78) Furthermore, in some embodiments, the antibodies or antigen-binding fragments thereof described herein can also contain one, two, or three heavy chain variable region CDRs selected from the group of SEQ ID NOs: 8-10, SEQ ID NOs: 22-24, SEQ ID NOs: 74-76, SEQ ID NOs: 34-36, SEQ ID NOs: 54-56, SEQ ID NOs: 64-66, and SEQ ID NOs: 44-46; and/or one, two, or three light chain variable region CDRs selected from the group of SEQ ID NOs: 5-7, SEQ ID NOs: 19-21, SEQ ID NOs: 71-73, SEQ ID NOs: 31-33, SEQ ID NOs: 51-53, SEQ ID NOs: 61-63, and SEQ ID Nos: 41-43.

(79) In some embodiments, the antibodies can have a heavy chain variable region (VH) comprising complementarity determining regions (CDRs) 1, 2, 3, wherein the CDR1 region comprises or consists of an amino acid sequence that is at least 80%, 85%, 90%, or 95% identical to a selected VH CDR1 amino acid sequence, the CDR2 region comprises or consists of an amino acid sequence that is at least 80%, 85%, 90%, or 95% identical to a selected VH CDR2 amino acid sequence, and the CDR3 region comprises or consists of an amino acid sequence that is at least 80%, 85%, 90%, or 95% identical to a selected VH CDR3 amino acid sequence. In some embodiments, the antibodies can have a light chain variable region (VL) comprising CDRs 1, 2, 3, wherein the CDR1 region comprises or consists of an amino acid sequence that is at least 80%, 85%, 90%, or 95% identical to a selected VL CDR1 amino acid sequence, the CDR2 region comprises or consists of an amino acid sequence that is at least 80%, 85%, 90%, or 95% identical to a selected VL CDR2 amino acid sequence, and the CDR3 region comprises or consists of an amino acid sequence that is at least 80%, 85%, 90%, or 95% identical to a selected VL CDR3 amino acid sequence. The selected VH CDRs 1, 2, 3 amino acid sequences and the selected VL CDRs, 1, 2, 3 amino acid sequences are shown in FIG. **9**.

(80) In some embodiments, the antibody or an antigen-binding fragment described herein can contain a heavy chain variable domain containing one, two, or three of the CDRs of SEQ ID NO: 8 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 9 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 10 with zero, one or two amino acid insertions, deletions, or substitutions.

(81) In some embodiments, the antibody or an antigen-binding fragment described herein can contain a heavy chain variable domain containing one, two, or three of the CDRs of SEQ ID NO: 22 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 23 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 24 with zero, one or two amino acid insertions, deletions, or substitutions.

(82) In some embodiments, the antibody or an antigen-binding fragment described herein can contain a heavy chain variable domain containing one, two, or three of the CDRs of SEQ ID NO:

74 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 75 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 76 with zero, one or two amino acid insertions, deletions, or substitutions.
(83) In some embodiments, the antibody or an antigen-binding fragment described herein can contain a heavy chain variable domain containing one, two, or three of the CDRs of SEQ ID NO: 34 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 35 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 36 with zero, one or two amino acid insertions, deletions, or substitutions.
(84) In some embodiments, the antibody or an antigen-binding fragment described herein can contain a light chain variable domain containing one, two, or three of the CDRs of SEQ ID NO: 5 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 6 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 7 with zero, one or two amino acid insertions, deletions, or substitutions.
(85) In some embodiments, the antibody or an antigen-binding fragment described herein can contain a light chain variable domain containing one, two, or three of the CDRs of SEQ ID NO: 19 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 20 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 21 with zero, one or two amino acid insertions, deletions, or substitutions.
(86) In some embodiments, the antibody or an antigen-binding fragment described herein can contain a light chain variable domain containing one, two, or three of the CDRs of SEQ ID NO: 71 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 72 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 73 with zero, one or two amino acid insertions, deletions, or substitutions.
(87) In some embodiments, the antibody or an antigen-binding fragment described herein can contain a light chain variable domain containing one, two, or three of the CDRs of SEQ ID NO: 31 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 32 with zero, one or two amino acid insertions, deletions, or substitutions; SEQ ID NO: 33 with zero, one or two amino acid insertions, deletions, or substitutions.
(88) The insertions, deletions, and substitutions can be within the CDR sequence, or at one or both terminal ends of the CDR sequence. In some embodiments, the CDR is determined based on Kabat numbering scheme.
(89) The disclosure also provides antibodies or antigen-binding fragments thereof that bind to PD-1. The antibodies or antigen-binding fragments thereof contain a heavy chain variable region (VH) comprising or consisting of an amino acid sequence that is at least 80%, 85%, 90%, or 95% identical to a selected VH sequence, and a light chain variable region (VL) comprising or consisting of an amino acid sequence that is at least 80%, 85%, 90%, or 95% identical to a selected VL sequence. In some embodiments, the selected VH sequence is SEQ ID NO: 4, and the selected VL sequence is SEQ ID NO: 2. In some embodiments, the selected VH sequence is SEQ ID NO: 18 and the selected VL sequence is SEQ ID NO: 16. In some embodiments, the selected VH sequence is SEQ ID NO: 70 and the selected VL sequence is SEQ ID NO: 68. In some embodiments, the selected VH sequence is SEQ ID NO: 30 and the selected VL sequence is SEQ ID NO: 28. In some embodiments, the selected VH sequence is SEQ ID NO: 50 and the selected VL sequence is SEQ ID NO: 48. In some embodiments, the selected VH sequence is SEQ ID NO: 60 and the selected VL sequence is SEQ ID NO: 58. In some embodiments, the selected VH sequence is SEQ ID NO: 40 and the selected VL sequence is SEQ ID NO: 38.
(90) The disclosure also provides nucleic acid comprising a polynucleotide encoding a polypeptide comprising an immunoglobulin heavy chain or an immunoglobulin light chain. The immunoglobulin heavy chain or immunoglobulin light chain comprises CDRs, or have sequences as shown in FIG. **9**. When the polypeptides are paired with corresponding polypeptide (e.g., a corresponding heavy chain variable region or a corresponding light chain variable region), the

paired polypeptides bind to PD-1 (e.g., human PD-1).
(91) The anti-PD-1 antibodies and antigen-binding fragments can also be antibody variants (including derivatives and conjugates) of antibodies or antibody fragments and multi-specific (e.g., bi-specific) antibodies or antibody fragments. Additional antibodies provided herein are polyclonal, monoclonal, multi-specific (multimeric, e.g., bi-specific), human antibodies, chimeric antibodies (e.g., human-mouse chimera), single-chain antibodies, intracellularly-made antibodies (i.e., intrabodies), and antigen-binding fragments thereof. The antibodies or antigen-binding fragments thereof can be of any type (e.g., IgG, IgE, IgM, IgD, IgA, and IgY), class (e.g., IgG1, IgG2, IgG3, IgG4, IgA1, and IgA2), or subclass. In some embodiments, the antibody or antigen-binding fragment thereof is an IgG antibody or antigen-binding fragment thereof.
(92) Fragments of antibodies are suitable for use in the methods provided so long as they retain the desired affinity and specificity of the full-length antibody. Thus, a fragment of an antibody that binds to PD-1 will retain an ability to bind to PD-1. An Fv fragment is an antibody fragment which contains a complete antigen recognition and binding site. This region consists of a dimer of one heavy and one light chain variable domain in tight association, which can be covalent in nature, for example in scFv. It is in this configuration that the three CDRs of each variable domain interact to define an antigen binding site on the surface of the VH-VL dimer. Collectively, the six CDRs or a subset thereof confer antigen binding specificity to the antibody. However, even a single variable domain (or half of an Fv comprising only three CDRs specific for an antigen) can have the ability to recognize and bind antigen, although usually at a lower affinity than the entire binding site. Single-chain Fv or (scFv) antibody fragments comprise the VH and VL domains (or regions) of antibody, wherein these domains are present in a single polypeptide chain. Generally, the scFv polypeptide further comprises a polypeptide linker between the VH and VL domains, which enables the scFv to form the desired structure for antigen binding.
(93) The present disclosure also provides an antibody or antigen-binding fragment thereof that cross-competes with any antibody or antigen-binding fragment as described herein. The cross-competing assay is known in the art, and is described e.g., in Moore et al., “Antibody cross-competition analysis of the human immunodeficiency virus type 1 gp120 exterior envelope glycoprotein.” Journal of virology 70.3 (1996): 1863-1872, which is incorporated herein reference in its entirety. In one aspect, the present disclosure also provides an antibody or antigen-binding fragment thereof that binds to the same epitope or region as any antibody or antigen-binding fragment as described herein. The epitope binning assay is known in the art, and is described e.g., in Estep et al. “High throughput solution-based measurement of antibody-antigen affinity and epitope binning.” MAbs. Vol. 5. No. 2. Taylor & Francis, 2013, which is incorporated herein reference in its entirety.
(94) In one aspect, this disclosure provides an isolated human PD-1 binding monoclonal antibody, or antigen-binding portion thereof comprising: (a) a light chain variable region comprising an amino acid sequence comprising SEQ ID NO:2; and (b) a heavy chain variable region comprising an amino acid sequence comprising SEQ ID NO:4; wherein the antibody or portion blocks the interaction between PD-1 and PD-L1 and also the interaction between PD-1 and PD-L2.
(95) In another aspect, this disclosure provides an isolated monoclonal antibody, or antigen-binding portion thereof comprising: (a) a light chain variable region comprising an amino acid sequence comprising SEQ ID NO:2; and (b) a heavy chain variable region comprising an amino acid sequence comprising SEQ ID NO:4; wherein the antibody specifically binds to PD-1 and blocks the interaction between PD-1 and PD-L1. In some embodiments, the monoclonal antibody, or an antigen-binding portion thereof stimulates an anti-tumor immune response. In some embodiments, the monoclonal antibody can be a chimeric antibody.
(96) In another aspect, this disclosure provides antibodies that comprise the heavy chain and light chain CDR1, CDR2 and CDR3 of clone #11. The amino acid sequences of the VL CDR1 of clone #11 is shown in SEQ ID NO:5. The amino acid sequences of the VL CDR2s of clone #11 is SEQ

ID NO:6. The amino acid sequences of the VL CDR3s of clone #11 is shown in SEQ ID NO:7. The amino acid sequence of the VH CDR1 of clone #11 is shown in SEQ ID NO:8. The amino acid sequence of the VH CDR2 of clone #11 is shown in SEQ ID NO:9. The amino acid sequence of the VH CDR3 of clone #11 is shown in SEQ ID NO:10. The CDR regions can be delineated using the Kabat system (Kabat et al. (1991) Sequences of Proteins of Immunological Interest, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242).

(97) In another aspect, this disclosure provides polynucleotide sequences encoding the light chain variable domain (VL) and heavy chain variable domains (VH) of the monoclonal antibody clone #11. The VL polynucleotide sequence is shown in SEQ ID NO:1. The VH polynucleotide sequence is shown in SEQ ID NO:3.

(98) Antibodies can be affinity maturated by light-chain shuffling combined with or without random mutagenesis of its heavy chain variable domain and panning against PD-1. The VL CDR1, CDR2 and CDR3 of the antibodies mentioned in this disclosure can be optimized with light-chain shuffling to create other anti-PD-1 binding molecules of the disclosure.

(99) An antibody of the disclosure further can be prepared using an antibody having one or more of the VH and/or VL sequences disclosed herein as starting material to engineer a modified antibody, which modified antibody can have altered properties from the starting antibody. An antibody can be engineered by modifying one or more residues within one or both variable regions (i.e., VH and/or VL), for example within one or more CDR regions and/or within one or more framework regions. Additionally, or alternatively, an antibody can be engineered by modifying residues within the constant region(s), for example to alter the effector function(s) of the antibody.

(100) In certain embodiments, CDR grafting can be used to engineer variable regions of antibodies. Antibodies interact with target antigens predominantly through amino acid residues that are located in the six heavy and light chain complementarity determining regions (CDRs).

(101) Because CDR sequences can be responsible for most antibody-antigen interactions, it can be possible to express recombinant antibodies that mimic the properties of specific naturally occurring antibodies by constructing expression vectors that include CDR sequences from the specific naturally occurring antibody grafted onto framework sequences from a different antibody with different properties (see, e.g., Riechmann et al. (1998) Nature 332:323-327; Jones et al. (1986) Nature 321: 522-525; Queen et al. (1989) Proc. Natl. Acad. See. U.S.A. 86: 10029-10033; U.S. Pat. Nos. 5,225,539; 5,530,101; 5,585,089; 5,693,762 and 6,180,370.)

(102) Accordingly, another embodiment of the disclosure pertains to an isolated monoclonal antibody, or antigen-binding portion thereof, comprising a light chain variable region comprising CDR1, CDR2, and CDR3 sequences comprising an amino acid sequence of SEQ ID NO:5, SEQ ID NO:6, and SEQ ID NO:7, respectively, and a heavy chain variable region a comprising CDR1, CDR2, and CDR3 sequences comprising an amino acid sequence of SEQ ID NO:8, SEQ ID NO:9 and SEQ ID NO:10, respectively. Thus, such antibodies contain the VH and VL CDR sequences of anti-PD-1 monoclonal antibody from hybridoma clone #11 and can contain different framework sequences from these antibodies.

(103) Such framework sequences can be obtained from public DNA databases or published references that include germline antibody gene sequences. For example, germline DNA sequences for human heavy and light chain variable region genes can be found in the “VBase” human germline sequence database (available on the Internet at www.mrc-cpe.cam.ac.uk/vbase), as well as in Kabat et al. (1991), cited supra; Tomlinson et al. (1992) “The Repertoire of Human Germline VH Sequences Reveals about Fifty Groups of VH Segments with Different Hypervariable Loops” J. Mol. Biol. 227:776-798; and Cox et al. (1994) “A Directory of Human Germ-line VH Segments Reveals a Strong Bias in their Usage” Eur. J. Immunol. 24:827-836; the contents of each of which are expressly incorporated herein by reference. As another example, the germline DNA sequences for human heavy and light chain variable region genes can be found in the GenBank database. For example, the following heavy chain germline sequences found in the HCo7 HuMAb mouse are

available in the accompanying GenBankAccession NOS.:1-69 (NG_0010109, NT_024637 & BC070333), 3-33 (NG_0010109 & NT_024637) and 3-7 (NG_0010109 &NT_024637). As another example, the following heavy chain germline sequences found in the HCo12 HuMAb mouse are available in the accompanying GenBankAccession NOS.: 1-69 (NG_0010109, NT_024637 & BC070333), 5-51 (NG_0010109 & NT_024637), 4-34 (NG_0010109 & NT_024637), 3-30.3 (CAJ556644) & 3-23 (AJ406678).

(104) Antibody protein sequences are compared against a compiled protein sequence database using one of the sequence similarity searching methods called the Gapped BLAST (Altschul et al. (1997)), which is well known to those skilled in the art. The compositions and methods of the presently disclosure are not limited to variants of the exemplary sequences disclosed herein but include those having at least 90%, at least 95% and at least 99% sequence identity to an exemplary sequence disclosed herein.

(105) Antibodies and Antigen Binding Fragments

(106) The present disclosure provides various antibodies and antigen-binding fragments thereof derived from anti-PD-1 antibodies described herein. In general, antibodies (also called immunoglobulins) are made up of two classes of polypeptide chains, light chains and heavy chains. A non-limiting examples of antibody of the present disclosure can be an intact, four immunoglobulin chain antibody comprising two heavy chains and two light chains. The heavy chain of the antibody can be of any isotype including IgM, IgG, IgE, IgA, or IgD or sub-isotype including IgG1, IgG2, IgG2a, IgG2b, IgG3, IgG4, IgE1, IgE2, etc. The light chain can be a kappa light chain or a lambda light chain. An antibody can comprise two identical copies of a light chain and two identical copies of a heavy chain. The heavy chains, which each contain one variable domain (or variable region, V.sub.H) and multiple constant domains (or constant regions), bind to one another via disulfide bonding within their constant domains to form the “stem” of the antibody. The light chains, which each contain one variable domain (or variable region, V.sub.L) and one constant domain (or constant region), each bind to one heavy chain via disulfide binding. The variable region of each light chain is aligned with the variable region of the heavy chain to which it is bound. The variable regions of both the light chains and heavy chains contain three hypervariable regions sandwiched between more conserved framework regions (FR).

(107) These hypervariable regions, known as the complementary determining regions (CDRs), form loops that comprise the antigen binding surface of the antibody. The four framework regions largely adopt a beta-sheet conformation and the CDRs form loops connecting the beta-sheet structure, and in some cases forming part of, the beta-sheet structure. The CDRs in each chain are held in close proximity by the framework regions and, with the CDRs from the other chain, contribute to the formation of the antigen-binding region.

(108) Methods for identifying the CDR regions of an antibody by analyzing the amino acid sequence of the antibody are well known, and a number of definitions of the CDRs are commonly used. The Kabat definition is based on sequence variability, and the Chothia definition is based on the location of the structural loop regions. These methods and definitions are described in, e.g., Martin, “Protein sequence and structure analysis of antibody variable domains,” Antibody engineering, Springer Berlin Heidelberg, 2001. 422-439; Abhinandan, et al. “Analysis and improvements to Kabat and structurally correct numbering of antibody variable domains,” Molecular immunology 45.14 (2008): 3832-3839; Wu, T.T. and Kabat, E.A. (1970) J. Exp. Med. 132: 211-250; Martin et al., Methods Enzymol. 203:121-53 (1991); Morea et al., Biophys Chem. 68(1-3):9-16 (October 1997); Morea et al., J Mol Biol. 275(2):269-94 (Jan. 1998); Chothia et al., Nature 342(6252):877-83 (December 1989); Ponomarenko and Bourne, BMC Structural Biology 7:64 (2007); each of which is incorporated herein by reference in its entirety. Unless otherwise indicated, Kabat definition is used in the present disclosure.

(109) The CDRs are important for recognizing an epitope of an antigen. As used herein, an “epitope” is the smallest portion of a target molecule capable of being specifically bound by the

antigen binding domain of an antibody. The minimal size of an epitope may be about three, four, five, six, or seven amino acids, but these amino acids need not be in a consecutive linear sequence of the antigen's primary structure, as the epitope may depend on an antigen's three-dimensional configuration based on the antigen's secondary and tertiary structure.

(110) In some embodiments, the antibody is an intact immunoglobulin molecule (e.g., IgG1, IgG2a, IgG2b, IgG3, IgM, IgD, IgE, IgA). The IgG subclasses (IgG1, IgG2, IgG3, and IgG4) are highly conserved, differ in their constant region, particularly in their hinges and upper CH2 domains. The sequences and differences of the IgG subclasses are known in the art, and are described, e.g., in Vidarsson, et al, "IgG subclasses and allotypes: from structure to effector functions." Frontiers in immunology 5 (2014); Irani, et al. "Molecular properties of human IgG subclasses and their implications for designing therapeutic monoclonal antibodies against infectious diseases." Molecular immunology 67.2 (2015): 171-182; Shakib, Farouk, ed. The human IgG subclasses: molecular analysis of structure, function and regulation. Elsevier, 2016; each of which is incorporated herein by reference in its entirety.

(111) The antibody can also be an immunoglobulin molecule that is derived from any species (e.g., human, rodent, mouse, camelid). Antibodies disclosed herein also include, but are not limited to, polyclonal, monoclonal, monospecific, polyspecific antibodies, and chimeric antibodies that include an immunoglobulin binding domain fused to another polypeptide. The term "antigen binding domain" or "antigen binding fragment" is a portion of an antibody that retains specific binding activity of the intact antibody, i.e., any portion of an antibody that is capable of specific binding to an epitope on the intact antibody's target molecule. It includes, e.g., Fab, Fab′, F(ab′)2, and variants of these fragments. Thus, in some embodiments, an antibody or an antigen binding fragment thereof can be, e.g., a scFv, a Fv, a Fd, a dAb, a bispecific antibody, a bispecific scFv, a diabody, a linear antibody, a single-chain antibody molecule, a multi-specific antibody formed from antibody fragments, and any polypeptide that includes a binding domain which is, or is homologous to, an antibody binding domain. Non-limiting examples of antigen binding domains include, e.g., the heavy chain and/or light chain CDRs of an intact antibody, the heavy and/or light chain variable regions of an intact antibody, full length heavy or light chains of an intact antibody, or an individual CDR from either the heavy chain or the light chain of an intact antibody.

(112) Fragments of antibodies are suitable for use in the methods described herein are also provided. The Fab fragment contains a variable and constant domain of the light chain and a variable domain and the first constant domain (CH1) of the heavy chain. F(ab′)2 antibody fragments comprise a pair of Fab fragments which are generally covalently linked near their carboxy termini by hinge cysteines between them. Other chemical couplings of antibody fragments are also known in the art.

(113) Diabodies are small antibody fragments with two antigen-binding sites, which fragments comprise a VH connected to a V.sub.L in the same polypeptide chain (VH and V.sub.L). By using a linker that is too short to allow pairing between the two domains on the same chain, the domains are forced to pair with the complementary domains of another chain and create two antigen-binding sites.

(114) Linear antibodies comprise a pair of tandem Fd segments (VH-CH1-VH-CH1) which, together with complementary light chain polypeptides, form a pair of antigen binding regions. Linear antibodies can be bispecific or monospecific.

(115) Antibodies and antibody fragments of the present disclosure can be modified in the Fc region to provide desired effector functions or serum half-life.

(116) Multimerization of antibodies may be accomplished through natural aggregation of antibodies or through chemical or recombinant linking techniques known in the art. For example, some percentage of purified antibody preparations (e.g., purified IgG.sub.1 molecules) spontaneously form protein aggregates containing antibody homodimers and other higher-order antibody multimers.

(117) Alternatively, antibody homodimers may be formed through chemical linkage techniques known in the art. For example, heterobifunctional crosslinking agents including, but not limited to SMCC (succinimidyl 4-(maleimidomethyl)cyclohexane-1-carboxylate) and SATA (N-succinimidyl S-acethylthio-acetate) can be used to form antibody multimers. An exemplary protocol for the formation of antibody homodimers is described in Ghetie et al. (*Proc. Natl. Acad. Sci. U.S.A.* 94: 7509-7514, 1997). Antibody homodimers can be converted to Fab′2 homodimers through digestion with pepsin. Another way to form antibody homodimers is through the use of the autophilic T15 peptide described in Zhao et al. (*J. Immunol.* 25:396-404, 2002).
(118) In some embodiments, the multi-specific antibody is a bi-specific antibody. Bi-specific antibodies can be made by engineering the interface between a pair of antibody molecules to maximize the percentage of heterodimers that are recovered from recombinant cell culture. For example, the interface can contain at least a part of the CH3 domain of an antibody constant domain. In this method, one or more small amino acid side chains from the interface of the first antibody molecule are replaced with larger side chains (e.g., tyrosine or tryptophan). Compensatory “cavities” of identical or similar size to the large side chain(s) are created on the interface of the second antibody molecule by replacing large amino acid side chains with smaller ones (e.g., alanine or threonine). This provides a mechanism for increasing the yield of the heterodimer over other unwanted end-products such as homodimers. This method is described, e.g., in WO 96/27011, which is incorporated by reference in its entirety.
(119) Bi-specific antibodies include cross-linked or “heteroconjugate” antibodies. For example, one of the antibodies in the heteroconjugate can be coupled to avidin and the other to biotin. Heteroconjugate antibodies can also be made using any convenient cross-linking methods. Suitable cross-linking agents and cross-linking techniques are well known in the art and are disclosed in U.S. Pat. No. 4,676,980, which is incorporated herein by reference in its entirety.
(120) Methods for generating bi-specific antibodies from antibody fragments are also known in the art. For example, bi-specific antibodies can be prepared using chemical linkage. Brennan et al. (Science 229:81, 1985) describes a procedure where intact antibodies are proteolytically cleaved to generate F(ab′).sub.2 fragments. These fragments are reduced in the presence of the dithiol complexing agent sodium arsenite to stabilize vicinal dithiols and prevent intermolecular disulfide formation. The Fab′ fragments generated are then converted to thionitrobenzoate (TNB) derivatives. One of the Fab′ TNB derivatives is then reconverted to the Fab′ thiol by reduction with mercaptoethylamine, and is mixed with an equimolar amount of another Fab′ TNB derivative to form the bi-specific antibody.
(121) Any of the antibodies or antigen-binding fragments described herein may be conjugated to a stabilizing molecule (e.g., a molecule that increases the half-life of the antibody or antigen-binding fragment thereof in a subject or in solution). Non-limiting examples of stabilizing molecules include: a polymer (e.g., a polyethylene glycol) or a protein (e.g., serum albumin, such as human serum albumin). The conjugation of a stabilizing molecule can increase the half-life or extend the biological activity of an antibody or an antigen-binding fragment in vitro (e.g., in tissue culture or when stored as a pharmaceutical composition) or in vivo (e.g., in a human).
(122) In some embodiments, the antibodies or antigen-binding fragments described herein can be conjugated to a therapeutic agent. The antibody-drug conjugate comprising the antibody or antigen-binding fragment thereof can covalently or non-covalently bind to a therapeutic agent. In some embodiments, the therapeutic agent is a cytotoxic or cytostatic agent (e.g., cytochalasin B, gramicidin D, ethidium bromide, emetine, mitomycin, etoposide, tenoposide, vincristine, vinblastine, colchicin, doxorubicin, daunorubicin, dihydroxy anthracin, maytansinoids such as DM-1 and DM-4, dione, mitoxantrone, mithramycin, actinomycin D, 1-dehydrotestosterone, glucocorticoids, procaine, tetracaine, lidocaine, propranolol, puromycin, epirubicin, and cyclophosphamide and analogs).
(123) In some embodiments, the antigen binding fragment can form a part of a chimeric antigen

receptor (CAR). In some embodiments, the chimeric antigen receptor are fusions of single-chain variable fragments (scFv) as described herein, fused to CD3-zeta transmembrane- and endodomain. In some embodiments, the chimeric antigen receptor also comprises intracellular signaling domains from various costimulatory protein receptors (e.g., CD28, 41BB, ICOS). In some embodiments, the chimeric antigen receptor comprises multiple signaling domains, e.g., CD3z-CD28-41BB or CD3z-CD28-OX40, to increase potency. Thus, in one aspect, the disclosure further provides cells (e.g., T cells) that express the chimeric antigen receptors as described herein.

(124) In some embodiments, the scFV has one heavy chain variable domain, and one light chain variable domain. In some embodiments, the scFV has two heavy chain variable domains, and two light chain variable domains.

(125) Antibody Characteristics

(126) The antibodies or antigen-binding fragments thereof described herein can block the binding between PD-1 and PD-1 ligands (e.g., PD-L1 or PD-L2). In some embodiments, by binding to PD-1, the antibody can inhibit PD-1 signaling pathway. In some embodiments, the antibody can upregulate immune response.

(127) In some embodiments, the antibodies or antigen-binding fragments thereof as described herein can increase immune response, activity or number of immune cells (e.g., T cells, CD8+ T cells, CD4+ T cells, macrophages, antigen presenting cells) by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 2 folds, 3 folds, 5 folds, 10 folds, or 20 folds.

(128) In some implementations, the antibody (or antigen-binding fragments thereof) specifically binds to PD-1 (e.g., human PD-1 (SEQ ID NO: 77), Rhesus monkey PD-1 (NP_001107830.1), mouse PD-1 (NP_032824.1), and/or chimeric PD-1) with a dissociation rate (koff) of less than 0.1 s.sup.−1, less than 0.01 s.sup.−1, less than 0.001 s.sup.−1, less than 0.0001 s.sup.−1, or less than 0.00001 s.sup.−1. In some embodiments, the dissociation rate (koff) is greater than 0.01 s.sup.−1, greater than 0.001 s.sup.−1, greater than 0.0001 s.sup.−1, greater than 0.00001 s.sup.−1, or greater than 0.000001 s.sup.−1.

(129) In some embodiments, kinetic association rates (kon) is greater than 1×10.sup.2/Ms, greater than 1×10.sup.3/Ms, greater than 1×10.sup.4/Ms, greater than 1×10.sup.5/Ms, or greater than 1×10.sup.6/Ms. In some embodiments, kinetic association rates (kon) is less than 1×10.sup.5/Ms, less than 1×10.sup.6/Ms, or less than 1×10.sup.7/Ms.

(130) Affinities can be deduced from the quotient of the kinetic rate constants (KD=koff/kon). In some embodiments, KD (Kd) is less than 1×10.sup.−6M, less than 1×10.sup.−7M, less than 1×10.sup.−8M, less than 1×10.sup.−9 M, or less than 1×10.sup.−10 M. In some embodiments, the KD is less than 50 nM, 30 nM, 20 nM, 15 nM, 10 nM, 9 nM, 8 nM, 7 nM, 6 nM, 5 nM, 4 nM, 3 nM, 2 nM, 1 nM, 0.9 nM, 0.8 nM, 0.7 nM, 0.6 nM, 0.5 nM, 0.4 nM, 0.3 nM, 0.2 nM, or 0.1 nM. In some embodiments, KD is greater than 1×10.sup.−7 M, greater than 1×10.sup.−8 M, greater than 1×10.sup.−9 M, greater than 1×10.sup.−10 M, greater than 1×10.sup.−11M, or greater than 1×10.sup.−12M.

(131) General techniques for measuring the affinity of an antibody for an antigen include, e.g., ELISA, RIA, and surface plasmon resonance (SPR). In some embodiments, the antibody binds to human PD-1 (SEQ ID NO: 77). In some embodiments, the antibody binds to monkey PD-1 (e.g., NP_001271065.1 from *Macaca fascicularis*), chimeric PD-1, and/or mouse PD-1 (e.g., NP_032824.1). In some embodiments, the antibody does not bind to monkey PD-1, chimeric PD-1, and/or mouse PD-1.

(132) In some embodiments, the antibodies or antigen-binding fragments thereof as described herein are PD-1 antagonist. In some embodiments, the antibodies or antigen binding fragments decrease PD-1 signal transduction in a target cell that expresses PD-1.

(133) In some embodiments, the antibodies or antigen binding fragments can enhance APC (e.g., DC cell) function, for example, inducing surface expression of costimulatory and MHC molecules, inducing production of proinflammatory cytokines, and/or enhancing T cell triggering function.

(134) In some embodiments, the Fc region is human IgG1, human IgG2, human IgG3, or human IgG4. In some embodiments, the antibody is a human IgG1 antibody.
(135) In some embodiments, the antibodies or antigen binding fragments do not have a functional Fc region. For example, the antibodies or antigen binding fragments are Fab, Fab′, F(ab′)2, and Fv fragments. In some embodiments, the Fc region has LALA mutations (L234A and L235A mutations in EU numbering), or LALA-PG mutations (L234A, L235A, P329G mutations in EU numbering).
(136) The binding of an antibody as described herein to PD-1 can also be assessed using one or more techniques well established in the art. For example, in a preferred embodiment, an antibody can be tested by ELISA assays, for example using a recombinant PD-1 protein. Still other suitable binding assays include but are not limited to a flow cytometry assay in which the antibody is reacted with a cell line that expresses human PD-1, such as HEK293T cells that have been transfected to express PD-1 (e.g., human PD-1) on their cell surface. Additionally or alternatively, the binding of the antibody, including the binding kinetics (e.g., KD value) can be tested in BIAcore binding assays and the like.
(137) Preferably, an antibody of the disclosure binds to a PD-1 protein with an EC50 of 5×10.sup.−8 M or less, binds to a PD-1 protein with a EC50 of 2×10.sup.−8M or less, binds to a PD-1 protein with a EC50 of 5×10.sup.−9 M or less, binds to a PD-1 protein with a EC50 of 4×10.sup.−9 M or less, binds to a PD-1 protein with a EC50 of 3×10.sup.−9 M or less, binds to a PD-1 protein with a EC50 of 2×10.sup.−9M or less, binds to a PD-1 protein with a EC50 of 1×10.sup.−9 M or less.
(138) Methods of Making Anti-PD-1 Antibodies
(139) An isolated fragment of human PD-1 can be used as an immunogen to generate antibodies using standard techniques for polyclonal and monoclonal antibody preparation. Polyclonal antibodies can be raised in animals by multiple injections (e.g., subcutaneous or intraperitoneal injections) of an antigenic peptide or protein. In some embodiments, the antigenic peptide or protein is injected with at least one adjuvant. In some embodiments, the antigenic peptide or protein can be conjugated to an agent that is immunogenic in the species to be immunized. Animals can be injected with the antigenic peptide or protein more than one time (e.g., twice, three times, or four times).
(140) The full-length polypeptide or protein can be used or, alternatively, antigenic peptide fragments thereof can be used as immunogens. The antigenic peptide of a protein comprises at least 8 (e.g., at least 10, 15, 20, or 30) amino acid residues of the amino acid sequence of PD-1 and encompasses an epitope of the protein such that an antibody raised against the peptide forms a specific immune complex with the protein. As described above, the full length sequence of human PD-1 is known in the art (SEQ ID NO: 77).
(141) An immunogen typically is used to prepare antibodies by immunizing a suitable subject (e.g., human or transgenic animal expressing at least one human immunoglobulin locus). An appropriate immunogenic preparation can contain, for example, a recombinantly-expressed or a chemically-synthesized polypeptide (e.g., a fragment of human PD-1). The preparation can further include an adjuvant, such as Freund's complete or incomplete adjuvant, or a similar immunostimulatory agent.
(142) Polyclonal antibodies can be prepared as described above by immunizing a suitable subject with a PD-1 polypeptide, or an antigenic peptide thereof (e.g., part of PD-1) as an immunogen. The antibody titer in the immunized subject can be monitored over time by standard techniques, such as with an enzyme-linked immunosorbent assay (ELISA) using the immobilized PD-1 polypeptide or peptide. If desired, the antibody molecules can be isolated from the mammal (e.g., from the blood) and further purified by well-known techniques, such as protein A of protein G chromatography to obtain the IgG fraction. At an appropriate time after immunization, e.g., when the specific antibody titers are highest, antibody-producing cells can be obtained from the subject and used to prepare monoclonal antibodies by standard techniques, such as the hybridoma technique originally

described by Kohler et al. (*Nature* 256:495-497, 1975), the human B cell hybridoma technique (Kozbor et al., *Immunol. Today* 4:72, 1983), the EBV-hybridoma technique (Cole et al., Monoclonal Antibodies and Cancer Therapy, Alan R. Liss, Inc., pp. 77-96, 1985), or trioma techniques. The technology for producing hybridomas is well known (see, generally, Current Protocols in Immunology, 1994, Coligan et al. (Eds.), John Wiley & Sons, Inc., New York, N.Y.). Hybridoma cells producing a monoclonal antibody are detected by screening the hybridoma culture supernatants for antibodies that bind the polypeptide or epitope of interest, e.g., using a standard ELISA assay.
(143) A nucleic acid molecule encoding the heavy or entire light chain of an anti-PD-1 antibody or portions thereof can be isolated from any source that produces such an antibody. In various embodiments, the nucleic acid molecules are isolated from a B cell isolated from an animal immunized with PD-1 or from an immortalized cell derived from such a B cell that expresses an anti-PD-1 antibody. Methods of isolating mRNA encoding an antibody are well-known in the art. See, e.g., Sambrook et al. The mRNA may be used to produce cDNA for use in the polymerase chain reaction (PCR) or cDNA cloning of antibody genes. In a preferred embodiment, the nucleic acid molecule is isolated from a hybridoma that has as one of its fusion partners a human immunoglobulin producing cell from a non-human transgenic animal. In another embodiment, the nucleic acid can be isolated from a non-human, nontransgenic animal. The nucleic acid molecules isolated from a non-human, nontransgenic animal may be used, e.g., for humanized antibodies.
(144) Variants of the antibodies or antigen-binding fragments described herein can be prepared by introducing appropriate nucleotide changes into the DNA encoding a human, humanized, or chimeric antibody, or antigen-binding fragment thereof described herein, or by peptide synthesis. Such variants include, for example, deletions, insertions, or substitutions of residues within the amino acids sequences that make-up the antigen-binding site of the antibody or an antigen-binding domain. In a population of such variants, some antibodies or antigen-binding fragments will have increased affinity for the target protein, e.g., PD-1. Any combination of deletions, insertions, and/or combinations can be made to arrive at an antibody or antigen-binding fragment thereof that has increased binding affinity for the target. The amino acid changes introduced into the antibody or antigen-binding fragment can also alter or introduce new post-translational modifications into the antibody or antigen-binding fragment, such as changing (e.g., increasing or decreasing) the number of glycosylation sites, changing the type of glycosylation site (e.g., changing the amino acid sequence such that a different sugar is attached by enzymes present in a cell), or introducing new glycosylation sites.
(145) Antibodies disclosed herein can be derived from any species of animal, including mammals. Non-limiting examples of native antibodies include antibodies derived from humans, primates, e.g., monkeys and apes, cows, pigs, horses, sheep, camelids (e.g., camels and llamas), chicken, goats, and rodents (e.g., rats, mice, hamsters and rabbits), including transgenic rodents genetically engineered to produce human antibodies.
(146) Human and humanized antibodies include antibodies having variable and constant regions derived from (or having the same amino acid sequence as those derived from) human germline immunoglobulin sequences. Human antibodies may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo), for example in the CDRs.
(147) A humanized antibody, typically has a human framework (FR) grafted with non-human CDRs. Thus, a humanized antibody has one or more amino acid sequence introduced into it from a source which is non-human. These non-human amino acid residues are often referred to as “import” residues, which are typically taken from an “import” variable domain.
(148) Ordinarily, amino acid sequence variants of the human, humanized, or chimeric anti-PD-1 antibody will contain an amino acid sequence having at least 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% percent identity with a sequence present in the light or heavy chain of the

original antibody.
(149) Identity or homology with respect to an original sequence is usually the percentage of amino acid residues present within the candidate sequence that are identical with a sequence present within the human, humanized, or chimeric anti-PD-1 antibody or fragment, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity.
(150) Additional modifications to the anti-PD-1 antibodies or antigen-binding fragments can be made. For example, a cysteine residue(s) can be introduced into the Fc region, thereby allowing interchain disulfide bond formation in this region. The homodimeric antibody thus generated may have any increased half-life in vitro and/or in vivo. Homodimeric antibodies with increased half-life in vitro and/or in vivo can also be prepared using heterobifunctional cross-linkers as described, for example, in Wolff et al. Wolff et al. ("Monoclonal antibody homodimers: enhanced antitumor activity in nude mice." Cancer research 53.11 (1993): 2560-2565). Alternatively, an antibody can be engineered which has dual Fc regions.
(151) In some embodiments, a covalent modification can be made to the anti-PD-1 antibody or antigen-binding fragment thereof. These covalent modifications can be made by chemical or enzymatic synthesis, or by enzymatic or chemical cleavage. Other types of covalent modifications of the antibody or antibody fragment are introduced into the molecule by reacting targeted amino acid residues of the antibody or fragment with an organic derivatization agent that is capable of reacting with selected side chains or the N- or C-terminal residues.
(152) In some embodiments, antibody variants are provided having a carbohydrate structure that lacks fucose attached (directly or indirectly) to an Fc region. For example, the amount of fucose in such antibody composition may be from 1% to 80%, from 1% to 65%, from 5% to 65% or from 20% to 40%. The amount of fucose is determined by calculating the average amount of fucose within the sugar chain at Asn297, relative to the sum of all glycostructures attached to Asn 297 (e.g. complex, hybrid and high mannose structures) as measured by MALDI-TOF mass spectrometry, as described in WO 2008/077546, for example. Asn297 refers to the asparagine residue located at about position 297 in the Fc region (EU numbering of Fc region residues; or position 314 in Kabat numbering); however, Asn297 may also be located about ±3 amino acids upstream or downstream of position 297, i.e., between positions 294 and 300, due to minor sequence variations in antibodies. Such fucosylation variants may have improved ADCC function. In some embodiments, to reduce glycan heterogeneity, the Fc region of the antibody can be further engineered to replace the Asparagine at position 297 with Alanine (N297A).
(153) In some embodiments, to facilitate production efficiency by avoiding Fab-arm exchange, the Fc region of the antibodies was further engineered to replace the serine at position 228 (EU numbering) of IgG4 with proline (S228P). A detailed description regarding S228 mutation is described, e.g., in Silva et al. "The S228P mutation prevents in vivo and in vitro IgG4 Fab-arm exchange as demonstrated using a combination of novel quantitative immunoassays and physiological matrix preparation." Journal of Biological Chemistry 290.9 (2015): 5462-5469, which is incorporated by reference in its entirety.
(154) Recombinant Vectors
(155) The present disclosure also provides recombinant vectors (e.g., an expression vectors) that include an isolated polynucleotide disclosed herein (e.g., a polynucleotide that encodes a polypeptide disclosed herein), host cells into which are introduced the recombinant vectors (i.e., such that the host cells contain the polynucleotide and/or a vector comprising the polynucleotide), and the production of recombinant antibody polypeptides or fragments thereof by recombinant techniques.
(156) As used herein, a "vector" is any construct capable of delivering one or more polynucleotide(s) of interest to a host cell when the vector is introduced to the host cell. An "expression vector" is capable of delivering and expressing the one or more polynucleotide(s) of

interest as an encoded polypeptide in a host cell into which the expression vector has been introduced. Thus, in an expression vector, the polynucleotide of interest is positioned for expression in the vector by being operably linked with regulatory elements such as a promoter, enhancer, and/or a poly-A tail, either within the vector or in the genome of the host cell at or near or flanking the integration site of the polynucleotide of interest such that the polynucleotide of interest will be translated in the host cell introduced with the expression vector.

(157) A vector can be introduced into the host cell by methods known in the art, e.g., electroporation, chemical transfection (e.g., DEAE-dextran), transformation, transfection, and infection and/or transduction (e.g., with recombinant virus). Thus, non-limiting examples of vectors include viral vectors (which can be used to generate recombinant virus), naked DNA or RNA, plasmids, cosmids, phage vectors, and DNA or RNA expression vectors associated with cationic condensing agents.

(158) In some implementations, a polynucleotide disclosed herein (e.g., a polynucleotide that encodes a polypeptide disclosed herein) is introduced using a viral expression system (e.g., vaccinia or other pox virus, retrovirus, or adenovirus), which may involve the use of a non-pathogenic (defective), replication competent virus, or may use a replication defective virus. In some embodiments, lentivirus is used to introduce the nucleic acids into the cells.

(159) For expression, the DNA insert comprising an antibody-encoding or polypeptide-encoding polynucleotide disclosed herein can be operatively linked to an appropriate promoter (e.g., a heterologous promoter), such as the phage lambda PL promoter, the *E. coli* lac, trp and tac promoters, the SV40 early and late promoters and promoters of retroviral LTRs, to name a few. Other suitable promoters are known to the skilled artisan. In some embodiments, the promoter is a cytomegalovirus (CMV) promoter. The expression constructs can further contain sites for transcription initiation, termination and, in the transcribed region, a ribosome binding site for translation. The coding portion of the mature transcripts expressed by the constructs may include a translation initiating at the beginning and a termination codon (UAA, UGA, or UAG) appropriately positioned at the end of the polypeptide to be translated.

(160) As indicated, the expression vectors can include at least one selectable marker. Such markers include dihydrofolate reductase or neomycin resistance for eukaryotic cell culture and tetracycline or ampicillin resistance genes for culturing in *E. coli* and other bacteria. Representative examples of appropriate hosts include, but are not limited to, bacterial cells, such as *E. coli*, *Streptomyces*, and *Salmonella typhimurium* cells; fungal cells, such as yeast cells; insect cells such as *Drosophila* S2 and *Spodoptera* Sf9 cells; animal cells such as CHO, COS, Bowes melanoma, and HK 293 cells; and plant cells. Appropriate culture mediums and conditions for the host cells described herein are known in the art.

(161) Introduction of the construct into the host cell can be effected by calcium phosphate transfection, DEAE-dextran mediated transfection, cationic lipid-mediated transfection, electroporation, transduction, infection or other methods. Such methods are described in many standard laboratory manuals, such as Davis et al., Basic Methods In Molecular Biology (1986), which is incorporated herein by reference in its entirety.

(162) Transcription of DNA encoding an antibody of the present disclosure by higher eukaryotes may be increased by inserting an enhancer sequence into the vector. Enhancers are cis-acting elements of DNA, usually about from 10 to 300 bp that act to increase transcriptional activity of a promoter in a given host cell-type. Examples of enhancers include the SV40 enhancer, which is located on the late side of the replication origin at base pairs 100 to 270, the cytomegalovirus early promoter enhancer, the polyoma enhancer on the late side of the replication origin, and adenovirus enhancers.

(163) For secretion of the translated protein into the lumen of the endoplasmic reticulum, into the periplasmic space or into the extracellular environment, appropriate secretion signals may be incorporated into the expressed polypeptide. The signals may be endogenous to the polypeptide or

they may be heterologous signals. In some embodiments, the signal peptide is SEQ ID NO: 78 or SEQ ID NO: 79.

(164) The polypeptide (e.g., antibody) can be expressed in a modified form, such as a fusion protein (e.g., a GST-fusion) or with a histidine-tag, and may include not only secretion signals, but also additional heterologous functional regions. For instance, a region of additional amino acids, particularly charged amino acids, may be added to the N-terminus of the polypeptide to improve stability and persistence in the host cell, during purification, or during subsequent handling and storage. Also, peptide moieties can be added to the polypeptide to facilitate purification. Such regions can be removed prior to final preparation of the polypeptide. The addition of peptide moieties to polypeptides to engender secretion or excretion, to improve stability and to facilitate purification, among others, are familiar and routine techniques in the art.

(165) In one aspect, the disclosure provides a nucleic acid that is at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% identical to SEQ ID NO: 1, 3, 15, 17, 27, 29, 37, 39, 47, 49, 47, 49, 67, or 69.

(166) The disclosure also provides a nucleic acid sequence that is at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% identical to any nucleotide sequence as described herein, and an amino acid sequence that is at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% identical to any amino acid sequence as described herein. In some embodiments, the disclosure relates to nucleotide sequences encoding any peptides that are described herein, or any amino acid sequences that are encoded by any nucleotide sequences as described herein. In some embodiments, the nucleic acid sequence is less than 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 150, 200, 250, 300, 350, 400, 500, or 600 nucleotides. In some embodiments, the amino acid sequence is less than 5, 6, 7, 8, 9, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, or 200 amino acid residues.

(167) In some embodiments, the amino acid sequence (i) comprises an amino acid sequence; or (ii) consists of an amino acid sequence, wherein the amino acid sequence is any one of the sequences as described herein. In some embodiments, the nucleic acid sequence (i) comprises a nucleic acid sequence; or (ii) consists of a nucleic acid sequence, wherein the nucleic acid sequence is any one of the sequences as described herein.

(168) To determine the percent identity of two amino acid sequences, or of two nucleic acid sequences, the sequences are aligned for optimal comparison purposes (e.g., gaps can be introduced in one or both of a first and a second amino acid or nucleic acid sequence for optimal alignment and non-homologous sequences can be disregarded for comparison purposes). The amino acid residues or nucleotides at corresponding amino acid positions or nucleotide positions are then compared. When a position in the first sequence is occupied by the same amino acid residue or nucleotide as the corresponding position in the second sequence, then the molecules are identical at that position. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences, taking into account the number of gaps, and the length of each gap, which need to be introduced for optimal alignment of the two sequences. For purposes of illustration, the comparison of sequences and determination of percent identity between two sequences can be accomplished using a Blossum 62 scoring matrix with a gap penalty of 12, a gap extend penalty of 4, and a frameshift gap penalty of 5.

(169) Methods of Treatment

(170) The antibodies, antibody compositions and methods of the present disclosure have numerous in vitro and in vivo utilities involving, for example, detection of PD-1 or enhancement of immune response by blockade of PD-1. In a preferred embodiment, the antibodies of the present disclosure are human antibodies. For example, these molecules can be administered to cells in culture, in vitro

or ex vivo, or to human subjects, e.g., in vivo, to enhance immunity in a variety of situations. Accordingly, in one aspect, the disclosure provides a method of modifying an immune response in a subject comprising administering to the subject the antibody, or antigen-binding portion thereof, of the disclosure such that the immune response in the subject is modified. Preferably, the response is enhanced, stimulated or up-regulated.

(171) As used herein, the term “subject” is intended to include human and non-human animals. Non-human animals include all vertebrates, e.g. mammals and non-mammals, such as non-human primates, sheep, dogs, cats, cows, horses, chickens, amphibians, and reptiles, although mammals are preferred, such as non-human primates, sheep, dogs, cats, cows and horses. Preferred subjects include human patients in need of enhancement of an immune response. The methods are particularly suitable for treating human patients having a disorder that can be treated by augmenting the T-cell mediated immune response. In a particular embodiment, the methods are particularly suitable for treatment of cancer cells in vivo. To achieve antigen-specific enhancement of immunity, the anti-PD-1 antibodies can be administered together with an antigen of interest. When antibodies to PD-1 are administered together with another agent, the two can be administered in either order or simultaneously.

(172) The disclosure further provides methods for detecting the presence of human PD-1 antigen in a sample, or measuring the amount of human PD-1 antigen, comprising contacting the sample, and a control sample, with a human monoclonal antibody, or an antigen-binding portion thereof, which specifically binds to human PD-1, under conditions that allow for formation of a complex between the antibody or portion thereof and human PD-1. The formation of a complex is then detected, wherein a difference complex formation between the sample compared to the control sample is indicative the presence of human PD-1 antigen in the sample.

(173) Blockade of PD-1 by antibodies can enhance the immune response to cancerous cells in the patient. In one aspect, the present disclosure relates to treatment of a subject in vivo using an anti-PD-1 antibody such that growth of cancerous tumors is inhibited. An anti-PD-1 antibody may be used alone to inhibit the growth of cancerous tumors. Alternatively, an anti-PD-1 antibody may be used in conjunction with other immunogenic agents, standard cancer treatments, or other antibodies, as described below.

(174) Accordingly, in one embodiment, the disclosure provides a method of inhibiting growth of tumor cells in a subject, comprising administering to the subject a therapeutically effective amount of an anti-PD-1 antibody, or antigen-binding portion thereof. Preferably, the antibody is a human anti-PD-1 antibody (such as any of the human anti-human PD-1 antibodies described herein). Additionally or alternatively, the antibody may be a chimeric or humanized anti-PD-1 antibody.

(175) Preferred cancers whose growth may be inhibited using the antibodies of the disclosure include cancers typically responsive to immunotherapy. Non-limiting examples of preferred cancers for treatment include melanoma (e.g., metastatic malignant melanoma), renal cancer (e.g. clear cell carcinoma), prostate cancer (e.g. hormone refractory prostate adenocarcinoma), breast cancer, colon cancer and lung cancer (e.g. non-small cell lung cancer). Additionally, the disclosure includes refractory or recurrent malignancies whose growth may be inhibited using the antibodies of the disclosure.

(176) Examples of other cancers that may be treated using the methods of the disclosure include bone cancer, pancreatic cancer, skin cancer, cancer of the head or neck, cutaneous or intraocular malignant melanoma, uterine cancer, ovarian cancer, rectal cancer, cancer of the anal region, stomach cancer, testicular cancer, uterine cancer, carcinoma of the fallopian tubes, carcinoma of the endometrium, carcinoma of the cervix, carcinoma of the vagina, carcinoma of the vulva, Hodgkin's Disease, non-Hodgkin's lymphoma, cancer of the esophagus, cancer of the small intestine, cancer of the endocrine system, cancer of the thyroid gland, cancer of the parathyroid gland, cancer of the adrenal gland, sarcoma of soft tissue, cancer of the urethra, cancer of the penis, chronic or acute leukemia including acute myeloid leukemia, chronic myeloid leukemia, acute lymphoblastic

leukemia, chronic lymphocytic leukemia, solid tumors of childhood, lymphocytic lymphoma, cancer of the bladder, cancer of the kidney or ureter, carcinoma of the renal pelvis, neoplasm of the central nervous system (CNS), primary CNS lymphoma, tumor angiogenesis, spinal axis tumor, brain stem glioma, pituitary adenoma, Kaposi's sarcoma, epidermoid cancer, squamous cell cancer, T-cell lymphoma, environmentally induced cancers including those induced by asbestos, and combinations of said cancers. The present disclosure is also useful for treatment of metastatic cancers, especially metastatic cancers that express PD-L1 (Iwai et al. (2005) Int. Immunol. 17:133-144).

(177) Optionally, antibodies to PD-1 can be combined with an immunogenic agent, such as cancerous cells, purified tumor antigens (including recombinant proteins, peptides, and carbohydrate molecules), cells, and cells transfected with genes encoding immune stimulating cytokines (He et al (2004) J. Immunol. 173:4919-28). Non-limiting examples of tumor vaccines that can be used include peptides of melanoma antigens, such as peptides of gp100, MAGE antigens, Trp-2, MARTI and/or tyrosinase, or tumor cells transfected to express the cytokine GM-CSF (discussed further below). In humans, some tumors have been shown to be immuno genie such as melanomas. It is anticipated that by raising the threshold of T cell activation by PD-1 blockade, we may expect to activate tumor responses in the host. PD-1 blockade is likely to be most effective when combined with a vaccination protocol. Many experimental strategies for vaccination against tumors have been devised (see Rosenberg, S., 2000, Development of Cancer Vaccines, ASCO Educational Book Spring: 60-62; Logothetis, C., 2000, ASCO Educational Book Spring: 300-302; Khayat, D. 2000, ASCO Educational Book Spring: 414-428; Poon, K. 2000, ASCO Educational Book Spring: 730-738; see also Restifo, N. and Sznol, M., Cancer Vaccines, Ch. 61, pp. 3023-3043 in De Vita, V. et al. (eds.), 1997, Cancer: Principles and Practice of Oncology. Fifth Edition). In one of these strategies, a vaccine is prepared using autologous or allogeneic tumor cells. These cellular vaccines have been shown to be most effective when the tumor cells are transduced to express GM-CSF. GM-CSF has been shown to be a potent activator of antigen presentation for tumor vaccination (Dranoff et al. (1993) Proc. Natl. Acad. Sci. U.S.A. 90: 3539-43).

(178) A PD-1 blockade may also be combined with standard cancer treatments. A PD-1 blockade may be effectively combined with chemotherapeutic regimes. In these instances, it may be possible to reduce the dose of chemotherapeutic reagent administered (Mokyr, M. et al. (1998) Cancer Research 58: 5301-5304). An example of such a combination is an anti-PD-1 antibody in combination with decarbazine for the treatment of melanoma. Another example of such a combination is an anti-PD-1 antibody in combination with interleukin-2 (IL-2) for the treatment of melanoma. The scientific rationale behind the combined use of PD-1 blockade and chemotherapy is that cell death, that is a consequence of the cytotoxic action of most chemotherapeutic compounds, should result in increased levels of tumor antigen in the antigen presentation pathway. Other combination therapies that may result in synergy with PD-1 blockade through cell death are radiation, surgery, and hormone deprivation. Each of these protocols creates a source of tumor antigen in the host. Angiogenesis inhibitors may also be combined with the PD-1 blockade. Inhibition of angiogenesis leads to tumor cell death which may feed tumor antigen into host antigen presentation pathways.

(179) PD-1 blocking antibodies can also be used in combination with bispecific antibodies that target Fc alpha or Fc gamma receptor-expressing effectors cells to tumor cells (see, e.g., U.S. Pat. Nos. 5,922,845 and 5,837,243). Bispecific antibodies can be used to target two separate antigens. For example anti-Fc receptor/anti-tumor antigen (e.g., Her-2/neu) bispecific antibodies have been used to target macrophages to sites of tumor. This targeting may more effectively activate tumor-specific responses. The T cell arm of these responses would be augmented by the use of PD-1 blockade. Alternatively, antigen may be delivered directly to DCs by the use of bispecific antibodies which bind to a tumor antigen and a dendritic cell-specific cell surface marker.

(180) Tumors evade host immune surveillance by a large variety of mechanisms. Many of these mechanisms may be overcome by the inactivation of proteins which are expressed by the tumors and which are immunosuppressive. These include among others TGF-beta (Kehrl, J. et al. (1986) J. Exp. Med. 163: 1037-1050), IL-10 (Howard, M. & O'Garra, A. (1992) Immunology Today 13: 198-200), and Fas ligand (Hahne, M. et al. (1996) Science 274: 1363-1365). Antibodies to each of these entities may be used in combination with anti-PD-1 antibodies of the present disclosure to counteract the effects of the immunosuppressive agent and favor tumor immune responses by the host.

(181) Other antibodies which may be used to activate host immune responsiveness can be used in combination with anti-PD-1. These include molecules on the surface of dendritic cells which activate DC function and antigen presentation. Anti-CD40 antibodies are able to substitute effectively for T cell helper activity (Ridge, J. et al. (1998) Nature 393:474-478) and can be used in conjunction with PD-1 antibodies (Ito, N. et al. (2000) Immunobiology 201 (5) 527-40). Activating antibodies to T cell costimulatory molecules such as CTLA-4 (e.g., U.S. Pat. No. 5,811,097), OX-40 (Weinberg, A. et al. (2000)Immunol 164: 2160-2169), 4-1BB (Melero, I. et al. (1997) Nature Medicine 3: 682-685 (1997), and ICOS (Hutloff, A. et al. (1999) Nature 397: 262-266) may also provide for increased levels of T cell activation.

(182) Bone marrow transplantation is currently being used to treat a variety of tumors of hematopoietic origin. While graft versus host disease is a consequence of this treatment, therapeutic benefit may be obtained from graft vs. tumor responses. PD-1 blockade can be used to increase the effectiveness of the donor engrafted tumor-specific T cells. There are also several experimental treatment protocols that involve ex vivo activation and expansion of antigen-specific T cells and adoptive transfer of these cells into recipients in order to antigen-specific T cells against tumor (Greenberg, R. & Riddell, S. (1999) Science 285: 546-51). These methods may also be used to activate T cell responses to infectious agents such as CMV. Ex vivo activation in the presence of anti-PD-1 antibodies may be expected to increase the frequency and activity of the adoptively transferred T cells.

(183) In one aspect, the disclosure provides methods for treating a cancer in a subject, methods of reducing the rate of the increase of volume of a tumor in a subject over time, methods of reducing the risk of developing a metastasis, or methods of reducing the risk of developing an additional metastasis in a subject. In some embodiments, the treatment can halt, slow, retard, or inhibit progression of a cancer. In some embodiments, the treatment can result in the reduction of in the number, severity, and/or duration of one or more symptoms of the cancer in a subject.

(184) In some embodiments, the antibody has a tumor growth inhibition percentage (TGI %) that is greater than 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 110%, 120%, 130%, 140%, 150%, 160%, 170%, 180%, 190%, or 200%. In some embodiments, the antibody has a tumor growth inhibition percentage that is less than 60%, 70%, 80%, 90%, 100%, 110%, 120%, 130%, 140%, 150%, 160%, 170%, 180%, 190%, or 200%. The TGI % can be determined, e.g., at 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 days after the treatment starts, or 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 months after the treatment starts. As used herein, the tumor growth inhibition percentage (TGI %) is calculated using the following formula:

$$\mathrm{TGI}\ (\%)=[1-(Ti-T0)/(Vi-V0)]\times 100$$

Ti is the average tumor volume in the treatment group on day i. T0 is the average tumor volume in the treatment group on day zero. Vi is the average tumor volume in the control group on day i. V0 is the average tumor volume in the control group on day zero.

(185) In some embodiments, the tumor inhibitory effects of the antibodies or antigen-binding fragments thereof as described herein are comparable to Pembrolizumab, Nivolumab, or Cemiplimab. In some embodiments, the tumor inhibitory effects of the antibodies or antigen-binding fragments thereof as described herein are at least 10%, 20%, 30%, 40%, 50%, 60%, 70%,

80%, 90%, 1 fold, 2 folds, or 5 folds more than Pembrolizumab, Nivolumab, or Cemiplimab.

(186) In one aspect, the disclosure features methods that include administering a therapeutically effective amount of an antibody or antigen-binding fragment thereof disclosed herein to a subject in need thereof (e.g., a subject having, or identified or diagnosed as having, a cancer), e.g., breast cancer (e.g., triple-negative breast cancer), carcinoid cancer, cervical cancer, endometrial cancer, glioma, head and neck cancer, liver cancer, lung cancer, small cell lung cancer, lymphoma, melanoma, ovarian cancer, pancreatic cancer, prostate cancer, renal cancer, colorectal cancer, gastric cancer, testicular cancer, thyroid cancer, bladder cancer, urethral cancer, or hematologic malignancy. In some embodiments, the cancer is unresectable melanoma or metastatic melanoma, non-small cell lung carcinoma (NSCLC), small cell lung cancer (SCLC), bladder cancer, or metastatic hormone-refractory prostate cancer. In some embodiments, the subject has a solid tumor. In some embodiments, the cancer is squamous cell carcinoma of the head and neck (SCCHN), renal cell carcinoma (RCC), triple-negative breast cancer (TNBC), or colorectal carcinoma. In some embodiments, the subject has Hodgkin's lymphoma. In some embodiments, the subject has triple-negative breast cancer (TNBC), gastric cancer, urothelial cancer, Merkel-cell carcinoma, or head and neck cancer. In some embodiments, the cancer is melanoma, pancreatic carcinoma, mesothelioma, hematological malignancies, especially Non-Hodgkin's lymphoma, lymphoma, chronic lymphocytic leukemia, or advanced solid tumors.

(187) In some embodiments, the compositions and methods disclosed herein can be used for treatment of patients at risk for a cancer. Patients with cancer can be identified with various methods known in the art.

(188) In one aspect, the disclosure provides methods for treating, preventing, or reducing the risk of developing disorders associated with an abnormal or unwanted immune response, e.g., an autoimmune disorder.

(189) As used herein, by an "effective amount" is meant an amount or dosage sufficient to effect beneficial or desired results including halting, slowing, retarding, or inhibiting progression of a disease, e.g., a cancer. An effective amount will vary depending upon, e.g., an age and a body weight of a subject to which the antibody, antigen binding fragment, antibody-encoding polynucleotide, vector comprising the polynucleotide, and/or compositions thereof is to be administered, a severity of symptoms and a route of administration, and thus administration can be determined on an individual basis.

(190) In some embodiments, one or more additional therapeutic agents can be administered to the subject. In some embodiments, the additional therapeutic agent can comprise one or more therapeutic agents selected from the group consisting of Trabectedin, nab-paclitaxel, Trebananib, Pazopanib, Cediranib, Palbociclib, everolimus, fluoropyrimidine, IFL, regorafenib, Reolysin, Alimta, Zykadia, Sutent, temsirolimus, axitinib, everolimus, sorafenib, Votrient, Pazopanib, IMA-901, AGS-003, cabozantinib, Vinflunine, an Hsp90 inhibitor, Ad-GM-CSF, Temazolomide, IL-2, IFNa, vinblastine, Thalomid, dacarbazine, cyclophosphamide, lenalidomide, azacytidine, lenalidomide, bortezomid, amrubicine, carfilzomib, pralatrexate, and enzastaurin.

(191) In some embodiments, the additional therapeutic agent can comprise one or more anti-cancer drugs (e.g., chemotherapies) selected from the group consisting of Abemaciclib, Abiraterone Acetate, Abraxane, Acalabrutinib, Actemra (Tocilizumab), Adcetris (Brentuximab Vedotin), Ado-Trastuzumab Emtansine, Adriamycin (Doxorubicin Hydrochloride), Afatinib Dimaleate, Afinitor (Everolimus), Akynzeo (Netupitant and Palonosetron Hydrochloride), Aldara (Imiquimod), Aldesleukin, Alecensa (Alectinib), Alectinib, Alemtuzumab, Alimta (Pemetrexed Disodium), Aliqopa (Copanlisib Hydrochloride), Alkeran, Aloxi (Palonosetron Hydrochloride), Alunbrig (Brigatinib), Ameluz (Aminolevulinic Acid Hydrochloride), Amifostine, Aminolevulinic Acid Hydrochloride, Anastrozole, Apalutamide, Aprepitant, Aranesp (Darbepoetin Alfa), Aredia (Pamidronate Disodium), Arimidex (Anastrozole), Aromasin (Exemestane), Arranon (Nelarabine), Arsenic Trioxide, Arzerra (Ofatumumab), Asparaginase *Erwinia chrysanthemi*, Asparlas

(Calaspargase Pegol-mknl), Atezolizumab, Avastin (Bevacizumab), Avelumab, Axicabtagene Ciloleucel, Axitinib, Azacitidine, Azedra (Iobenguane I 131), Bavencio (Avelumab), Beleodaq (Belinostat), Belinostat, Bendamustine Hydrochloride, Bendeka (Bendamustine Hydrochloride), Besponsa (Inotuzumab Ozogamicin), Bevacizumab, Bexarotene, Bicalutamide, BiCNU (Carmustine), Binimetinib, Bleomycin Sulfate, Blinatumomab, Blincyto (Blinatumomab), Bortezomib, Bosulif (Bosutinib), Bosutinib, Braftovi (Encorafenib), Brentuximab Vedotin, Brigatinib, BuMel, Busulfan, Busulfex (Busulfan), Cabazitaxel, Cablivi (Caplacizumab-yhdp), Cabometyx (Cabozantinib-S-Malate), Cabozantinib-S-Malate, Calaspargase Pegol-mknl, Calquence (Acalabrutinib), Campath (Alemtuzumab), Camptosar (Irinotecan Hydrochloride), Capecitabine, Caplacizumab-yhdp, Carac (Fluorouracil—Topical), Carboplatin, CARBOPLATIN-TAXOL, Carfilzomib, Carmustine, Carmustine Implant, Casodex (Bicalutamide), Cemiplimab-rwlc, Ceritinib, Cerubidine (Daunorubicin Hydrochloride), Cervarix (Recombinant HPV Bivalent Vaccine), Cetuximab, Chlorambucil, CHLORAMBUCIL-PREDNISONE, Cisplatin, Cladribine, Clofarabine, Clolar (Clofarabine), Cobimetinib, Cometriq (Cabozantinib-S-Malate), Copanlisib Hydrochloride, Copiktra (Duvelisib), Cosmegen (Dactinomycin), Cotellic (Cobimetinib), Crizotinib, Cyclophosphamide, Cyramza (Ramucirumab), Cytarabine, Cytarabine Liposome, Cytosar-U (Cytarabine), Dabrafenib, Dacarbazine, Dacogen (Decitabine), Dacomitinib, Dactinomycin, Daratumumab, Darbepoetin Alfa, Darzalex (Daratumumab), Dasatinib, Daunorubicin Hydrochloride, Daunorubicin Hydrochloride and Cytarabine Liposome, Daurismo (Glasdegib Maleate), Decitabine, Defibrotide Sodium, Defitelio (Defibrotide Sodium), Degarelix, Denileukin Diftitox, Denosumab, DepoCyt (Cytarabine Liposome), Dexamethasone, Dexrazoxane Hydrochloride, Dinutuximab, Docetaxel, Doxil (Doxorubicin Hydrochloride Liposome), Doxorubicin Hydrochloride, Doxorubicin Hydrochloride Liposome, Dox-SL (Doxorubicin Hydrochloride Liposome), Durvalumab, Duvelisib, Efudex (Fluorouracil—Topical), Eligard (Leuprolide Acetate), Elitek (Rasburicase), Ellence (Epirubicin Hydrochloride), Elotuzumab, Eloxatin (Oxaliplatin), Eltrombopag Olamine, Elzonris (Tagraxofusp-erzs), Emapalumab-lzsg, Emend (Aprepitant), Empliciti (Elotuzumab), Enasidenib Mesylate, Encorafenib, Enzalutamide, Epirubicin Hydrochloride, Epoetin Alfa, Epogen (Epoetin Alfa), Erbitux (Cetuximab), Eribulin Mesylate, Erivedge (Vismodegib), Erleada (Apalutamide), Erlotinib Hydrochloride, Erwinaze (Asparaginase *Erwinia chrysanthemi*), Ethyol (Amifostine), Etopophos (Etoposide Phosphate), Etoposide, Etoposide Phosphate, Evacet (Doxorubicin Hydrochloride Liposome), Everolimus, Evista (Raloxifene Hydrochloride), Evomela (Melphalan Hydrochloride), Exemestane, 5-FU (Fluorouracil), Fareston (Toremifene), Farydak (Panobinostat), Faslodex (Fulvestrant), Femara (Letrozole), Filgrastim, Firmagon (Degarelix), Fludarabine Phosphate, Fluoroplex (Fluorouracil—Topical), Fluorouracil Injection, Fluorouracil—Topical, Flutamide, Folotyn (Pralatrexate), Fostamatinib Disodium, Fulvestrant, Fusilev (Leucovorin Calcium), Gamifant (Emapalumab-lzsg), Gardasil (Recombinant HPV Quadrivalent Vaccine), Gardasil 9 (Recombinant HPV Nonavalent Vaccine), Gazyva (Obinutuzumab), Gefitinib, Gemcitabine Hydrochloride, GEMCITABINE-CISPLATIN, GEMCITABINE-OXALIPLATIN, Gemtuzumab Ozogamicin, Gemzar (Gemcitabine Hydrochloride), Gilotrif (Afatinib Dimaleate), Gilteritinib Fumarate, Glasdegib Maleate, Gleevec (Imatinib Mesylate), Gliadel Wafer (Carmustine Implant), Glucarpidase, Goserelin Acetate, Granisetron, Granisetron Hydrochloride, Granix (Filgrastim), Halaven (Eribulin Mesylate), Hemangeol (Propranolol Hydrochloride), Herceptin Hylecta (Trastuzumab and Hyaluronidase-oysk), Herceptin (Trastuzumab), HPV Bivalent Vaccine, Recombinant, HPV Nonavalent Vaccine, Recombinant, HPV Quadrivalent Vaccine, Recombinant, Hycamtin (Topotecan Hydrochloride), Hydrea (Hydroxyurea), Hydroxyurea, Hyper-CVAD, Ibrance (Palbociclib), Ibritumomab Tiuxetan, Ibrutinib, Iclusig (Ponatinib Hydrochloride), Idarubicin Hydrochloride, Idelalisib, Idhifa (Enasidenib Mesylate), Ifex (Ifosfamide), Ifosfamide, IL-2 (Aldesleukin), Imatinib Mesylate, Imbruvica (Ibrutinib), Imfinzi (Durvalumab), Imiquimod, Imlygic (Talimogene Laherparepvec), Inlyta (Axitinib), Inotuzumab Ozogamicin, Interferon Alfa-2b, Recombinant, Interleukin-2

(Aldesleukin), Intron A (Recombinant Interferon Alfa-2b), Iobenguane I 131, Ipilimumab, Iressa (Gefitinib), Irinotecan Hydrochloride, Irinotecan Hydrochloride Liposome, Istodax (Romidepsin), Ivosidenib, Ixabepilone, Ixazomib Citrate, Ixempra (Ixabepilone), Jakafi (Ruxolitinib Phosphate), Jevtana (Cabazitaxel), Kadcyla (Ado-Trastuzumab Emtansine), Kepivance (Palifermin), Keytruda (Pembrolizumab), Kisqali (Ribociclib), Kymriah (Tisagenlecleucel), Kyprolis (Carfilzomib), Lanreotide Acetate, Lapatinib Ditosylate, Larotrectinib Sulfate, Lartruvo (Olaratumab), Lenalidomide, Lenvatinib Mesylate, Lenvima (Lenvatinib Mesylate), Letrozole, Leucovorin Calcium, Leukeran (Chlorambucil), Leuprolide Acetate, Levulan Kerastik (Aminolevulinic Acid Hydrochloride), Libtayo (Cemiplimab-rwlc), LipoDox (Doxorubicin Hydrochloride Liposome), Lomustine, Lonsurf (Trifluridine and Tipiracil Hydrochloride), Lorbrena (Lorlatinib), Lorlatinib, Lumoxiti (Moxetumomab Pasudotox-tdfk), Lupron (Leuprolide Acetate), Lupron Depot (Leuprolide Acetate), Lutathera (Lutetium Lu 177-Dotatate), Lutetium (Lu 177-Dotatate), Lynparza (Olaparib), Marqibo (Vincristine Sulfate Liposome), Matulane (Procarbazine Hydrochloride), Mechlorethamine Hydrochloride, Megestrol Acetate, Mekinist (Trametinib), Mektovi (Binimetinib), Melphalan, Melphalan Hydrochloride, Mercaptopurine, Mesna, Mesnex (Mesna), Methotrexate, Methylnaltrexone Bromide, Midostaurin, Mitomycin C, Mitoxantrone Hydrochloride, Mogamulizumab-kpkc, Moxetumomab Pasudotox-tdfk, Mozobil (Plerixafor), Mustargen (Mechlorethamine Hydrochloride), Myleran (Busulfan), Mylotarg (Gemtuzumab Ozogamicin), Nanoparticle Paclitaxel (Paclitaxel Albumin-stabilized Nanoparticle Formulation), Navelbine (Vinorelbine Tartrate), Necitumumab, Nelarabine, Neratinib Maleate, Nerlynx (Neratinib Maleate), Netupitant and Palonosetron Hydrochloride, Neulasta (Pegfilgrastim), Neupogen (Filgrastim), Nexavar (Sorafenib Tosylate), Nilandron (Nilutamide), Nilotinib, Nilutamide, Ninlaro (Ixazomib Citrate), Niraparib Tosylate Monohydrate, Nivolumab, Nplate (Romiplostim), Obinutuzumab, Odomzo (Sonidegib), Ofatumumab, Olaparib, Olaratumab, Omacetaxine Mepesuccinate, Oncaspar (Pegaspargase), Ondansetron Hydrochloride, Onivyde (Irinotecan Hydrochloride Liposome), Ontak (Denileukin Diftitox), Opdivo (Nivolumab), Osimertinib, Oxaliplatin, Paclitaxel, Paclitaxel Albumin-stabilized Nanoparticle Formulation, Palbociclib, Palifermin, Palonosetron Hydrochloride, Palonosetron Hydrochloride and Netupitant, Pamidronate Disodium, Panitumumab, Panobinostat, Pazopanib Hydrochloride, Pegaspargase, Pegfilgrastim, Peginterferon Alfa-2b, PEG-Intron (Peginterferon Alfa-2b), Pembrolizumab, Pemetrexed Disodium, Perjeta (Pertuzumab), Pertuzumab, Plerixafor, Pomalidomide, Pomalyst (Pomalidomide), Ponatinib Hydrochloride, Portrazza (Necitumumab), Poteligeo (Mogamulizumab-kpkc), Pralatrexate, Prednisone, Procarbazine Hydrochloride, Procrit (Epoetin Alfa), Proleukin (Aldesleukin), Prolia (Denosumab), Promacta (Eltrombopag Olamine), Propranolol Hydrochloride, Provenge (Sipuleucel-T), Purinethol (Mercaptopurine), Purixan (Mercaptopurine), Radium 223 Dichloride, Raloxifene Hydrochloride, Ramucirumab, Rasburicase, Ravulizumab-cwvz, Recombinant Human Papillomavirus (HPV) Bivalent Vaccine, Recombinant Human Papillomavirus (HPV) Nonavalent Vaccine, Recombinant Human Papillomavirus (HPV) Quadrivalent Vaccine, Recombinant Interferon Alfa-2b, Regorafenib, Relistor (Methylnaltrexone Bromide), Retacrit (Epoetin Alfa), Revlimid (Lenalidomide), Rheumatrex (Methotrexate), Ribociclib, Rittman (Rituximab), Rituxan Hycela (Rituximab and Hyaluronidase Human), Rituximab, Rituximab and Hyaluronidase Human, Rolapitant Hydrochloride, Romidepsin, Romiplostim, Rubidomycin (Daunorubicin Hydrochloride), Rubraca (Rucaparib Camsylate), Rucaparib Camsylate, Ruxolitinib Phosphate, Rydapt (Midostaurin), Sancuso (Granisetron), Sclerosol Intrapleural Aerosol (Talc), Siltuximab, Sipuleucel-T, Somatuline Depot (Lanreotide Acetate), Sonidegib, Sorafenib Tosylate, Sprycel (Dasatinib), Sterile Talc Powder (Talc), Steritalc (Talc), Stivarga (Regorafenib), Sunitinib Malate, Sustol (Granisetron), Sutent (Sunitinib Malate), Sylatron (Peginterferon Alfa-2b), Sylvant (Siltuximab), Synribo (Omacetaxine Mepesuccinate), Tabloid (Thioguanine), Tafinlar (Dabrafenib), Tagraxofusp-erzs, Tagrisso (Osimertinib), Talazoparib Tosylate, Talimogene Laherparepvec, Talzenna (Talazoparib Tosylate), Tamoxifen

Citrate, Tarabine PFS (Cytarabine), Tarceva (Erlotinib Hydrochloride), Targretin (Bexarotene), Tasigna (Nilotinib), Tavalisse (Fostamatinib Disodium), Taxol (Paclitaxel), Taxotere (Docetaxel), Tecentriq (Atezolizumab), Temodar (Temozolomide), Temozolomide, Temsirolimus, Thalidomide, Thalomid (Thalidomide), Thioguanine, Thiotepa, Tibsovo (Ivosidenib), Tisagenlecleucel, Tocilizumab, Tolak (Fluorouracil—Topical), Topotecan Hydrochloride, Toremifene, Torisel (Temsirolimus), Totect (Dexrazoxane Hydrochloride), Trabectedin, Trametinib, Trastuzumab, Trastuzumab and Hyaluronidase-oysk, Treanda (Bendamustine Hydrochloride), Trexall (Methotrexate), Trifluridine and Tipiracil Hydrochloride, Trisenox (Arsenic Trioxide), Tykerb (Lapatinib Ditosylate), Ultomiris (Ravulizumab-cwvz), Unituxin (Dinutuximab), Uridine Triacetate, Valrubicin, Valstar (Valrubicin), Vandetanib, VAMP, Varubi (Rolapitant Hydrochloride), Vectibix (Panitumumab), VeIP, Velcade (Bortezomib), Vemurafenib, Venclexta (Venetoclax), Venetoclax, Verzenio (Abemaciclib), Vidaza (Azacitidine), Vinblastine Sulfate, Vincristine Sulfate, Vincristine Sulfate Liposome, Vinorelbine Tartrate, Vismodegib, Vistogard (Uridine Triacetate), Vitrakvi (Larotrectinib Sulfate), Vizimpro (Dacomitinib), Voraxaze (Glucarpidase), Vorinostat, Votrient (Pazopanib Hydrochloride), Vyxeos (Daunorubicin Hydrochloride and Cytarabine Liposome), Xalkori (Crizotinib), Xeloda (Capecitabine), Xgeva (Denosumab), Xofigo (Radium 223 Dichloride), Xospata (Gilteritinib Fumarate), Xtandi (Enzalutamide), Yervoy (Ipilimumab), Yescarta (Axicabtagene Ciloleucel), Yondelis (Trabectedin), Zaltrap (Ziv-Aflibercept), Zarxio (Filgrastim), Zejula (Niraparib Tosylate Monohydrate), Zelboraf (Vemurafenib), Zevalin (Ibritumomab Tiuxetan), Zinecard (Dexrazoxane Hydrochloride), Ziv-Aflibercept, Zofran (Ondansetron Hydrochloride), Zoladex (Goserelin Acetate), Zoledronic Acid, Zolinza (Vorinostat), Zometa (Zoledronic Acid), Zydelig (Idelalisib), Zykadia (Ceritinib), or Zytiga (Abiraterone Acetate). Many of these anti-cancer drugs, including their dosage information, are described on the National Cancer Institute website, which is incorporated by reference in their entirety.

(192) In some embodiments, the chemotherapeutic agent is 5-fluorouracil, bleomycin, capecitabine, cisplatin, cyclophosphamide, dacarbazine, doxorubicin, etoposide, folinic acid, methotrexate, oxaliplatin, prednisolone, procarbazine, vinblastine, vinorelbine, docetaxel, epirubicin, or mustine.

(193) In some embodiments, the method as described herein can increase the efficacy of the anti-cancer drug (e.g., chemotherapy) by at least 50%, 60%, 70%, 80%, 90%, 1 fold, 2 folds, 3 folds, or 5 folds (e.g., as compared to the efficacy when only the anti-cancer drug is administered to the subject at the same dose). Because the efficacy of the anti-cancer drug has been improved, in some embodiments, the therapeutically effective dose of the anti-cancer drug (e.g., chemotherapeutic agent) is about or at least 10%, 20%, 30%, 40%, or 50% lower than a typical dose (e.g., an FDA-approved dose) for the anti-cancer drug, thereby minimizing side effects. In addition, because the efficacy of the anti-cancer drug has been improved, in some embodiments, the therapeutically effective dose of the anti-cancer drug can be administered to the subject less frequently. For example, during a treatment period, the interval between administrations can be increased by about or at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 1 fold, 2 folds, 3 folds, or 5 folds than the administration interval in an FDA-approved treatment plan for the anti-cancer drug. In some embodiments, the total number of administrations can be reduced in a treatment period. For example, the total number of administrations can be reduced by about or at least 10%, 20%, 30%, 40%, or 50%. In some embodiments, because of the improved efficacy, the length of the treatment period can be shortened, e.g., by about or at least 10%, 20%, 30%, 40%, or 50% than the length of an FDA-approved treatment plan.

(194) In some embodiments, the additional therapeutic agent can be adoptive cell therapies. Adoptive cell therapies include a type of immunotherapy in which T cells (a type of immune cell) are given to a patient to help the body fight diseases, such as cancer. In cancer therapy, T cells are usually taken from the patient's own blood or tumor tissue, grown in large numbers in the laboratory, and then given back to the patient to help the immune system fight the cancer. In some embodiments, the adoptive cell therapies include chimeric antigen receptor T-cell (CAR T-cell)

therapy, tumor-infiltrating lymphocyte (TIL) therapy, engineered TCR therapy, or natural killer (NK) cell therapy. In some embodiments, the adoptive cell therapies can be called adoptive cell transfer, cellular adoptive immunotherapy, or T-cell transfer therapy.
(195) In some embodiments, the additional therapeutic agent can comprise one or more therapeutic agents selected from the group consisting of an adjuvant, a TLR agonist, tumor necrosis factor (TNF) alpha, IL-1, HMGB1, an IL-10 antagonist, an IL-4 antagonist, an IL-13 antagonist, an IL-17 antagonist, an HVEM antagonist, an ICOS agonist, a treatment targeting CX3CL1, a treatment targeting CXCL9, a treatment targeting CXCL10, a treatment targeting CCL5, an LFA-1 agonist, an ICAM1 agonist, and a Selectin agonist.
(196) In some embodiments, carboplatin, nab-paclitaxel, paclitaxel, cisplatin, pemetrexed, gemcitabine, FOLFOX, or FOLFIRI are administered to the subject.
(197) In some embodiments, the additional therapeutic agent is an anti-OX40 antibody, an anti-PD-1 antibody, an anti-PD-L1 antibody, an anti-PD-L2 antibody, an anti-LAG-3 antibody, an anti-TIGIT antibody, an anti-BTLA antibody, an anti-CTLA-4 antibody, or an anti-GITR antibody.
(198) Treating Infectious Diseases
(199) The disclosure also provides methods of treating patients that have been exposed to toxins or pathogens. Accordingly, another aspect of the disclosure provides a method of treating an infectious disease in a subject, comprising administering to the subject an anti-PD-1 antibody, or antigen-binding portion thereof, such that the subject is treated for the infectious disease. Preferably, the antibody is a human anti-human PD-1 antibody (such as any of the human anti-PD-1 antibodies described herein). Additionally or alternatively, the antibody can be a chimeric or humanized antibody.
(200) Similar to the use as discussed above, antibody mediated PD-1 blockade can be used alone, or as an adjuvant, in combination with vaccines, to stimulate the immune response to pathogens, toxins, and self-antigens. Examples of pathogens for which this therapeutic approach can be used, include, e.g., pathogens for which there is currently no effective vaccine, or pathogens for which conventional vaccines are not completely effective. These include, but are not limited to HIV, Hepatitis (A, B, or C), Influenza, Herpes, Giardia, Malaria, *Leishmania*, *Staphylococcus aureus*, *Pseudomonas Aeruginosa*. The PD-1 blockade is particularly useful against established infections by pathogens such as HIV that present mutated antigens over the course of the infections. These new epitopes can be recognized by the immune system when the anti-human PD-1 antibody is administered to the subject, thus provoking a strong T cell response that is not dampened by negative signals through PD-1.
(201) Some examples of pathogenic viruses causing infections treatable by methods of the disclosure include HIV, hepatitis (A, B, or C), herpes virus (e.g., VZV, HSV-1, HAV-6, HSY-II, and CMY, Epstein Barr virus), adenovirus, influenza virus, flaviviruses, echovirus, rhinovirus, coxsackie virus, coronavirus, respiratory syncytial virus, mumps virus, rotavirus, measles virus, rubella virus, parvovirus, vaccinia virus, HTLV virus, dengue virus, papillomavirus, molluscum virus, poliovirus, rabies virus, JC virus and arboviral encephalitis virus.
(202) Some examples of pathogenic bacteria causing infections treatable by methods of the disclosure include *chlamydia*, *rickettsia* bacteria, mycobacteria, staphylococci, streptococci, pneumonococci, meningococci and gonococci, *klebsiella*, *proteus*, *serratia*, *pseudomonas*, *legionella*, diphtheria, *salmonella*, bacilli, cholera, tetanus, botulism, anthrax, plague, leptospirosis, and Lymes disease bacteria.
(203) Some examples of pathogenic fungi causing infections treatable by methods of the disclosure include *Candida* (*albicans*, *krusei*, *glabrata*, *tropicalis*, etc.), *Cryptococcus neoformans*, *Aspergillus* (*fumigatus*, *niger*, etc.), Genus Mucorales (*mucor*, *absidia*, rhizophus), *Sporothrix schenkii*, *Blastomyces dermatitidis*, *Paracoccidioides brasiliensis*, *Coccidioides immitis* and *Histoplasma capsulatum*.
(204) Some examples of pathogenic parasites causing infections treatable by methods of the

disclosure include *Entamoeba histolytica, Balantidium coli, Naegleriafowleri, Acanthamoeba* sp., *Giardia Zambia, Cryptosporidium* sp., *Pneumocystis carinii, Plasmodium vivax, Babesia microti, Trypanosoma brucei, Trypanosoma cruzi, Leishmania donovani, Toxoplasma gondi*, and *Nippostrongylus brasiliensis.*

(205) In all of the above methods, PD-1 blockade can be combined with other forms of immunotherapy such as cytokine treatment (e.g., interferons, GM-CSF, G-CSF, IL-2), or bispecific antibody therapy, which provides for enhanced presentation of tumor antigens (see, e.g., Bolliger (1993) Proc. Natl. Acad. Sci. USA 90:6444-6448; Poljak (1994) Structure 2:1121-1123).

(206) Bispecific antibodies or antigen-binding fragments can be produced by a variety of methods including fusion of hybridomas or linking of Fab′ fragments. See, e.g., Songsivilai & Lachmarm, Clin. Exp. Immunol. 79: 315-321 (1990), Kostelny et al., J. Immunol. 148:1547-1553 (1992). In addition, bispecific antibodies can be formed as “diabodies” or “Janusins.” In some embodiments, the bispecific antibody binds to two different epitopes of PD-1. In some embodiments, the bispecific antibody has a first heavy chain and a first light chain from a monoclonal antibody of hybridoma clone #11, and an additional antibody heavy chain and light chain. In some embodiments, the additional light chain and heavy chain also are from one of the above-identified monoclonal antibodies, but can be different from the first heavy and light chains.

(207) Pharmaceutical Formulations

(208) In another aspect, the present disclosure provides a composition, e.g., a pharmaceutical composition, containing one or a combination of monoclonal antibodies, or antigen-binding portion(s) thereof, of the present disclosure, formulated together with a pharmaceutically acceptable carrier. Such compositions may include one or a combination of (e.g., two or more different) antibodies, or immunoconjugates or bispecific molecules of the disclosure. For example, a pharmaceutical composition of the disclosure can comprise a combination of antibodies (or immunoconjugates or bispecific antibodies) that bind to different epitopes on the target antigen or that have complementary activities.

(209) Pharmaceutical compositions of the disclosure also can be administered in combination therapy, i.e., combined with other agents. For example, the combination therapy can include an anti-PD-1 antibody of the present disclosure combined with at least one other anti-inflammatory or immunosuppressant agent. Examples of therapeutic agents that can be used in combination therapy are described in greater detail below in the section on uses of the antibodies of the disclosure.

(210) A “pharmaceutically acceptable carrier” refers to an ingredient in a pharmaceutical formulation, other than an active ingredient, which is nontoxic to a subject. A pharmaceutically acceptable carrier includes, but is not limited to, a buffer, excipient, stabilizer, or preservative.

(211) The pharmaceutical compositions of the disclosure may include one or more pharmaceutically acceptable salts. A “pharmaceutically acceptable salt” refers to a salt that retains the desired biological activity of the parent compound and does not impart any undesired toxicological effects (see e.g., Berge, S. M., et al. (1977) J. Pharm. Sci. 66:1-19). Examples of such salts include acid addition salts and base addition salts. Acid addition salts include those derived from nontoxic inorganic acids, such as hydrochloric, nitric, phosphoric, sulfuric, hydrobromic, hydroiodic, phosphorous and the like, as well as from nontoxic organic acids such as aliphatic mono and dicarboxylic acids, phenyl-substituted alkanoic acids, hydroxy alkanoic acids, aromatic acids, aliphatic and aromatic sulfonic acids and the like. Base addition salts include those derived from alkaline earth metals, such as sodium, potassium, magnesium, calcium and the like, as well as from nontoxic organic amines, such as N,N′-dibenzylethylenediamine, N-methylglucamine, chloroprocaine, choline, diethanolamine, ethylenediamine, procaine and the like.

(212) A pharmaceutical composition of the disclosure also may include a pharmaceutically acceptable anti-oxidant. Examples of pharmaceutically acceptable antioxidants include: (1) water soluble antioxidants, such as ascorbic acid, cysteine hydrochloride, sodium bisulfate, sodium metabisulfite, sodium sulfite and the like; (2) oil-soluble antioxidants, such as ascorbyl palmitate,

butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT), lecithin, propyl gallate, alpha-tocopherol, and the like; and (3) metal chelating agents, such as citric acid, ethylenediamine tetraacetic acid (EDTA), sorbitol, tartaric acid, phosphoric acid, and the like.

(213) Examples of suitable aqueous and nonaqueous carriers that may be employed in the pharmaceutical compositions of the disclosure include water, ethanol, polyols (such as glycerol, propylene glycol, polyethylene glycol, and the like), and suitable mixtures thereof, vegetable oils, such as olive oil, and injectable organic esters, such as ethyl oleate. Proper fluidity can be maintained, for example, by the use of coating materials, such as lecithin, by the maintenance of the required particle size in the case of dispersions, and by the use of surfactants.

(214) These compositions may also contain adjuvants such as preservatives, wetting agents, emulsifying agents and dispersing agents. Prevention of presence of microorganisms may be ensured both by sterilization procedures, supra, and by the inclusion of various antibacterial and antifungal agents, for example, paraben, chlorobutanol, phenol sorbic acid, and the like. It may also be desirable to include isotonic agents, such as sugars, sodium chloride, and the like into the compositions. In addition, prolonged absorption of the injectable pharmaceutical form may be brought about by the inclusion of agents which delay absorption such as aluminum monostearate and gelatin.

(215) Pharmaceutically acceptable carriers include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersion. The use of such media and agents for pharmaceutically active substances is known in the art. Except insofar as any conventional media or agent is incompatible with the active compound, use thereof in the pharmaceutical compositions of the disclosure is contemplated. Supplementary active compounds can also be incorporated into the compositions. Therapeutic compositions typically must be sterile and stable under the conditions of manufacture and storage. The composition can be formulated as a solution, microemulsion, liposome, or other ordered structure suitable to high drug concentration. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), and suitable mixtures thereof. The proper fluidity can be maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. In many cases, it will be preferable to include isotonic agents, for example, sugars, polyalcohols such as mannitol, sorbitol, or sodium chloride in the composition. Prolonged absorption of the injectable compositions can be brought about by including in the composition an agent that delays absorption, for example, monostearate salts and gelatin.

(216) Sterile injectable solutions can be prepared by incorporating the active compound in the required amount in an appropriate solvent with one or a combination of ingredients enumerated above, as required, followed by sterilization microfiltration. Generally, dispersions are prepared by incorporating the active compound into a sterile vehicle that contains a basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum drying and freeze-drying (lyophilization) that yield a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof.

(217) The amount of active ingredient which can be combined with a carrier material to produce a single dosage form will vary depending upon the subject being treated, and the particular mode of administration. The amount of active ingredient which can be combined with a carrier material to produce a single dosage form will generally be that amount of the composition which produces a therapeutic effect. Generally, out of one hundred percent, this amount will range from about 0.01 percent to about ninety-nine percent of active ingredient, preferably from about 0.1 percent to about 70 percent, most preferably from about 1 percent to about 30 percent of active ingredient in combination with a pharmaceutically acceptable carrier.

(218) Dosage regimens are adjusted to provide the optimum desired response (e.g., a therapeutic response). For example, a single bolus may be administered, several divided doses may be administered over time or the dose may be proportionally reduced or increased as indicated by the exigencies of the therapeutic situation. It is especially advantageous to formulate parenteral compositions in dosage unit form for ease of administration and uniformity of dosage. Dosage unit form as used herein refers to physically discrete units suited as unitary dosages for the subjects to be treated; each unit contains a predetermined quantity of active compound calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier. The specification for the dosage unit forms of the disclosure are dictated by and directly dependent on (a) the unique characteristics of the active compound and the particular therapeutic effect to be achieved, and (b) the limitations inherent in the art of compounding such an active compound for the treatment of sensitivity in individuals.
(219) For administration of the antibody, the dosage ranges from about 0.0001 to 100 mg/kg, and more usually 0.01 to 5 mg/kg, of the host body weight. For example dosages can be 0.3 mg/kg body weight, 1 mg/kg body weight, 3 mg/kg body weight, 5 mg/kg body weight or 10 mg/kg body weight or within the range of 1-10 mg/kg. An exemplary treatment regime entails administration once per week, once every two weeks, once every three weeks, once every four weeks, once a month, once every 3 months or once every three to 6 months. Preferred dosage regimens for an anti-PD-1 antibody of the disclosure include 1 mg/kg body weight or 3 mg/kg body weight via intravenous administration, with the antibody being given using one of the following dosing schedules: (i) every four weeks for six dosages, then every three months; (ii) every three weeks; (iii) 3 mg/kg body weight once followed by 1 mg/kg body weight every three weeks.
(220) In some methods, two or more monoclonal antibodies with different binding specificities are administered simultaneously, in which case the dosage of each antibody administered falls within the ranges indicated. Antibody is usually administered on multiple occasions. Intervals between single dosages can be, for example, weekly, monthly, every three months or yearly. Intervals can also be irregular as indicated by measuring blood levels of antibody to the target antigen in the patient. In some methods, dosage is adjusted to achieve a plasma antibody concentration of about 1-1000 μg/ml and in some methods about 25-300 μg/ml.
(221) Alternatively, antibody can be administered as a sustained release formulation, in which case less frequent administration is required. Dosage and frequency vary depending on the half-life of the antibody in the patient. In general, human antibodies show the longest half-life, followed by humanized antibodies, chimeric antibodies, and nonhuman antibodies. The dosage and frequency of administration can vary depending on whether the treatment is prophylactic or therapeutic. In prophylactic applications, a relatively low dosage is administered at relatively infrequent intervals over a long period of time. Some patients continue to receive treatment for the rest of their lives. In therapeutic applications, a relatively high dosage at relatively short intervals is sometimes required until progression of the disease is reduced or terminated, and preferably until the patient shows partial or complete amelioration of symptoms of disease. Thereafter, the patient can be administered a prophylactic regime.
(222) Actual dosage levels of the active ingredients in the pharmaceutical compositions of the present disclosure may be varied so as to obtain an amount of the active ingredient which is effective to achieve the desired therapeutic response for a particular patient, composition, and mode of administration, without being toxic to the patient. The selected dosage level will depend upon a variety of pharmacokinetic factors including the activity of the particular compositions of the present disclosure employed, or the ester, salt or amide thereof, the route of administration, the time of administration, the rate of excretion of the particular compound being employed, the duration of the treatment, other drugs, compounds and/or materials used in combination with the particular compositions employed, the age, sex, weight, condition, general health and prior medical history of the patient being treated, and like factors well known in the medical arts.

(223) A “therapeutically effective dosage” of an anti-PD-1 antibody of the disclosure preferably results in a decrease in severity of disease symptoms, an increase in frequency and duration of disease symptom-free periods, or a prevention of impairment or disability due to the disease affliction. For example, for the treatment of tumors, a “therapeutically effective dosage” preferably inhibits cell growth or tumor growth by at least about 20%, more preferably by at least about 40%, even more preferably by at least about 60%, and still more preferably by at least about 80% relative to untreated subjects. The ability of a compound to inhibit tumor growth can be evaluated in an animal model system predictive of efficacy in human tumors. Alternatively, this property of a composition can be evaluated by examining the ability of the compound to inhibit, such inhibition in vitro by assays known to the skilled practitioner. A therapeutically effective amount of a therapeutic compound can decrease tumor size, or otherwise ameliorate symptoms in a subject. One of ordinary skill in the art would be able to determine such amounts based on such factors as the subject's size, the severity of the subject's symptoms, and the particular composition or route of administration selected.
(224) In another aspect, the instant disclosure provides a pharmaceutical kit of parts comprising an anti-PD-1 antibody and an anti-CTLA-4 antibody, as described herein. The kit may also further comprise instructions for use in the treatment of a hyperproliferative disease (such as cancer as described herein). In another embodiment, the anti-PD-1 and anti-CTLA-4 antibodies may be co-packaged in unit dosage form.
(225) In certain embodiments, two or more monoclonal antibodies with different binding specificities (e.g., anti-PD-1 and anti-CTLA-4) are administered simultaneously, in which case the dosage of each antibody administered falls within the ranges indicated. Antibody can be administered as a single dose or more commonly can be administered on multiple occasions. Intervals between single dosages can be, for example, weekly, monthly, every three months or yearly. Intervals can also be irregular as indicated by measuring blood levels of antibody to the target antigen in the patient. In some methods, dosage is adjusted to achieve a plasma antibody concentration of about 1-1000 μg/ml and in some methods about 25-300 μg/ml.
(226) A composition of the present disclosure can be administered via one or more routes of administration using one or more of a variety of methods known in the art. As will be appreciated by the skilled artisan, the route and/or mode of administration will vary depending upon the desired results. Preferred routes of administration for antibodies of the disclosure include intravenous, intramuscular, intradermal, intraperitoneal, subcutaneous, spinal or other parenteral routes of administration, for example by injection or infusion. The phrase “parenteral administration” as used herein means modes of administration other than enteral and topical administration, usually by injection, and includes, without limitation, intravenous, intramuscular, intraarterial, intrathecal, intracapsular, intraorbital, intracardiac, intradermal, intraperitoneal, transtracheal, subcutaneous, subcuticular, intraarticular, subcapsular, subarachnoid, intraspinal, epidural and intrasternal injection and infusion.
(227) Alternatively, an antibody of the disclosure can be administered via a non-parenteral route, such as a topical, epidermal or mucosal route of administration, for example, intranasally, orally, vaginally, rectally, sublingually or topically.
(228) The active compounds can be prepared with carriers that will protect the compound against rapid release, such as a controlled release formulation, including implants, transdermal patches, and microencapsulated delivery systems. Biodegradable, biocompatible polymers can be used, such as ethylene vinyl acetate, polyanhydrides, polyglycolic acid, collagen, polyorthoesters, and poly lactic acid. Many methods for the preparation of such formulations are patented or generally known to those skilled in the art. See, e.g., Sustained and Controlled Release Drug Delivery Systems, J. R. Robinson, ed., Marcel Dekker, Inc., New York, 1978.
(229) Therapeutic compositions can be administered with medical devices known in the art. For example, in a preferred embodiment, a therapeutic composition of the disclosure can be

administered with a needleless hypodermic injection device, such as the devices disclosed in U.S. Pat. Nos. 5,399,163; 5,383,851; 5,312,335; 5,064,413; 4,941,880; 4,790,824; or 4,596,556. Examples of well-known implants and modules useful in the present disclosure include: U.S. Pat. No. 4,487,603, which discloses an implantable micro-infusion pump for dispensing medication at a controlled rate; U.S. Pat. No. 4,486,194, which discloses a therapeutic device for administering medicants through the skin; U.S. Pat. No. 4,447,233, which discloses a medication infusion pump for delivering medication at a precise infusion rate; U.S. Pat. No. 4,447,224, which discloses a variable flow implantable infusion apparatus for continuous drug delivery; U.S. Pat. No. 4,439,196, which discloses an osmotic drug delivery system having multi-chamber compartments; and U.S. Pat. No. 4,475,196, which discloses an osmotic drug delivery system. These patents are incorporated herein by reference. Many other such implants, delivery systems, and modules are known to those skilled in the art.

(230) In certain embodiments, the human monoclonal antibodies of the disclosure can be formulated to ensure proper distribution in vivo. For example, the blood-brain barrier (BBB) excludes many highly hydrophilic compounds. To ensure that the therapeutic compounds of the disclosure cross the BBB (if desired), they can be formulated, for example, in liposomes. For methods of manufacturing liposomes, see, e.g., U.S. Pat. Nos. 4,522,811; 5,374,548; and 5,399,331. The liposomes may comprise one or more moieties which are selectively transported into specific cells or organs, thus enhance targeted drug delivery (see, e.g., V. V. Ranade (1989) J. Clin. Pharmacol. 29:685). Exemplary targeting moieties include folate or biotin (see, e.g., U.S. Pat. No. 5,416,016 to Low et al.); mannosides (Umezawa et al., (1988) Biochem. Biophys. Res. Commun. 153:1038); antibodies (P. G. Bloeman et al. (1995) FEES Lett. 357: 140; M. Owais et al. (1995) Antimicrob. Agents Chemother. 39: 180); surfactant protein A receptor (Briscoe et al. (1995) Am. J. Physiol. 1233:134); p120 (Schreier et al. (1994) J. Biol. Chem. 269:9090); see also K. Keinanen; M. L. Laukkanen (1994)FEBS Lett. 346:123; J. J. Killion; I. J. Fidler (1994) Immunomethods 4:273.

(231) Also provided herein are pharmaceutical compositions that contain at least one (e.g., one, two, three, or four) of the antibodies or antigen-binding fragments described herein. Two or more (e.g., two, three, or four) of any of the antibodies or antigen-binding fragments described herein can be present in a pharmaceutical composition in any combination. The pharmaceutical compositions may be formulated in any manner known in the art.

(232) The pharmaceutical compositions can be included in a container, pack, or dispenser together with instructions for administration. The disclosure also provides methods of manufacturing the antibodies or antigen binding fragments thereof for various uses as described herein.

EXAMPLES

(233) The invention is further described in the following examples, which do not limit the scope of the invention described in the claims.

Example 1. Generating Mouse Anti-PD-1 Antibodies

(234) Experiments were performed to generate mouse antibodies against human PD-1. Mice were immunized with human PD-1. Spleen cells were collected. Hybridomas were screened for binding affinities. A hybridoma (clone #11) was selected for further experiments. The sequences of the antibodies were determined by sequencing (F02).

(235) Western Blot analysis was performed with the anti-PD-1 antibody obtained from the selected hybridoma (clone #11) (FIG. **1**). HEK-293T cells were purchased from American Type Culture Collection (ATCC). HEK-293T cells were transfected with the pCMV6-ENTRY control (Left lane) or pCMV6-ENTRY with PDCD1 cDNA (Origene, Cat #RC210364; Right lane) for 48 hours and lysed. Equivalent amounts of cell lysates (5 μg per lane) were separated by SDS-PAGE and immunoblotted with the anti-PD-1 antibodies (1:2000). The results show that the anti-PD-1 antibodies can bind to human PD-1 with high affinity.

Example 2. Immunofluorescent Staining

(236) Immunofluorescent staining was performed on cells expressing human PD-1. 293T-PD-1 cells were produced by lentiviral transduction of HEK-293T cells with a vector expressing human PD-1. Cells were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% Fetal bovine serum (FBS).
(237) FIG. **2**A shows the immunofluorescent staining of PDCD1 on RC210364-stable-transfected HEK293T cells with mouse monoclonal antibody anti-PD-1. The nucleus was labeled with Hoechst33342. The same experiments were performed on HEK293T cells as a negative control (1:100) (FIG. **2**B).
(238) FIG. **3**A shows immunocytochemistry staining of stable expression PD1 cells using anti-PD-1 mouse monoclonal antibody. FIG. **3**B shows immunocytochemistry staining of 293T cells as a negative control (1:900).
(239) Flow cytometric analysis was also performed on cells expressing PD-1 using anti-PD-1 antibody and a nonspecific control antibody (1:50) (FIG. **4**).
(240) FIG. **5** shows flow cytometric analysis of cells with stable expression of PD-L1 (Origene, Cat #RC213071) using anti-PD-1 antibody from hybridoma clone #11 (F02) or 0.3 ug/ml PD1-Fc fusion protein (TP700199) or both, and detected by anti-Fc (human) IgG-FITC (1:50). Binding of PD-L1 to PD-1 was completely blocked by anti-PD-1 antibody from hybridoma clone #11. The result shows that the anti-PD-1 antibody can effectively block the binding of PD-L1 to PD-1.
(241) FIG. **6** shows flow cytometric analysis of PD-L2(RC224141) transiently transfected HEK293T cells using anti-PD-1 antibody from hybridoma clone #11 or 1 μg/ml PD1-Fc fusion protein (TP700199) or both, and detected by anti-Fc (human) IgG-FITC (1:50). Binding of PD-L2 to PD-1 was completely blocked by anti-PD-1 antibody from hybridoma clone #11. The result shows that the anti-PD-1 antibody can effectively block the binding of PD-L2 to PD-1.
Example 3. Humanization of Antibodies
(242) In general, antibody specific sequences (CDRs) from the mouse parent antibody were grafted onto human donor sequences. The donor sequences were selected according to bioinformatics software that utilizes extensive antibody database to calculate a unique humanness score for each combination. These antibodies were cloned into an expression vector and transfected into cells lines for recombinant protein expression. Antibody affinity to the target protein, PD-1, was tested in vitro by ELISA and cell-based binding assays.
(243) The humanization of antibodies was achieved by CDR grafting and resurfacing strategies. Moreover, de-immunization strategy was also used. Bioinformatics tools including antibody modelling and critical framework residues identification were utilized to design humanized heavy and light chains. Those with the maximal humanization scores were combined to obtain full antibodies. Many of these antibodies will have an affinity that is comprisable to the original antibody.
(244) The humanized VH and VL were cloned into transient expression vectors. The final constructs were confirmed by sequencing. Further, the candidate pairs (including one chimeric pair for control) of heavy and light chain were co-transfected into CHO or HEK293 cells for transient expression. The expressed antibodies were purified, followed by measurement of their epitope specificity and affinity by ELISA or cell based-binding assay.
Example 4. Binding Affinity of Anti-PD1 to Recombinant Human PD-1 (rhPD-1)
(245) 5 μg/ml of anti-PD1 antibody in 100 μl coating buffer was coated in the microwells at 4° C. overnight. By following the ELISA procedure, a 5-fold serial dilution of recombinant biotinylated human PD-1 was added to the assay plate pre-coated with anti-PD1 antibodies. The bound human PD-1 protein was then detected with Sav-HRP reagent. The results were analyzed through Prism 7 and Kd values were determined by the program.
(246) As shown in FIGS. **7**A-**7**G, most of the anti-PD1 antibodies could bind to the recombinant human PD-1 with a Kd value, which was determined by ELISA, at nM range. For example, the estimated Kd for h11 is 0.29 nM, the estimated Kd for Ab7 is 0.68 nM, and the estimated Kd for

Ab8 is 0.32 nM. The hl 1, Ab7 and Ab8 antibodies exhibited comparable Kd values to the original mouse clone F02.

Example 5. Binding Affinity of Anti-PD1 to Human PD-1 (hPD-1) on Cell Surface

(247) 293T-PD-1 cells (0.03×10.sup.6/well) were incubated with different concentrations of each tested anti-PD1 antibody at 4° C. for 60 minutes. The cells were then stained with a secondary antibody (APC-anti-Hu IgG Fc for humanized antibodies and FITC-anti-Mouse IgG F(ab′)2 for F02) for 30 minutes (1:50), followed by flow cytometric analysis. The results were analyzed through Prism 7 and Kd values were determined by the program.

(248) As shown in FIGS. **8**A-**8**G, all of the anti-PD1 antibodies can bind to the PD-1 protein expressed on the 293T-PD-1 cells. The Kd values, determined by flow cytometry, for all the antibodies were at nM range. For example, the estimated Kd for h11 is 0.55 nM, the estimated Kd for Ab2 is 3.24 nM, the estimated Kd for Ab4 is 41 nM, the estimated Kd for Ab5 is 3.69 nM, the estimated Kd for Ab7 is 2.3 nM, and the estimated Kd for Ab8 is 0.65 nM.

OTHER EMBODIMENTS

(249) It is to be understood that while the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

Claims

1. An antibody or antigen-binding fragment thereof that binds to PD-1 (Programed Cell Death Protein 1) comprising: a heavy chain variable region (VH) comprising complementarity determining regions (CDRs) 1, 2, and 3, wherein the VH CDR1 region comprises an amino acid sequence that is identical to a selected VH CDR1 amino acid sequence, the VH CDR2 region comprises an amino acid sequence that is identical to a selected VH CDR2 amino acid sequence, and the VH CDR3 region comprises an amino acid sequence that is % identical to a selected VH CDR3 amino acid sequence; and a light chain variable region (VL) comprising CDRs 1, 2, and 3, wherein the VL CDR1 region comprises an amino acid sequence that is identical to a selected VL CDR1 amino acid sequence, the VL CDR2 region comprises an amino acid sequence that is identical to a selected VL CDR2 amino acid sequence, and the VL CDR3 region comprises an amino acid sequence that is identical to a selected VL CDR3 amino acid sequence, wherein the selected VH CDRs 1, 2, and 3 amino acid sequences and the selected VL CDRs, 1, 2, and 3 amino acid sequences are one of the following: (1) the selected VH CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 8, 9, 10, respectively, and the selected VL CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 5, 6, 7, respectively; (2) the selected VH CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 22, 23, 24, respectively, and the selected VL CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 19, 20, 21, respectively; (3) the selected VH CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 74, 75, 76, respectively, and the selected VL CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 71, 72, 73, respectively; and (4) the selected VH CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 34, 35, 36, respectively, and the selected VL CDRs 1, 2, 3 amino acid sequences are set forth in SEQ ID NOs: 31, 32, 33, respectively.

2. The antibody or antigen-binding fragment thereof of claim 1, wherein the VH comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 8, 9, and 10, respectively, and the VL comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 5, 6, and 7, respectively.

3. The antibody or antigen-binding fragment thereof of claim 1, wherein the VH comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 22, 23, and 24, respectively, and the VL comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 19, 20,

and 21, respectively.

4. The antibody or antigen-binding fragment thereof of claim 1, wherein the VH comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 74, 75, and 76, respectively, and the VL comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 71, 72, and 73, respectively.

5. The antibody or antigen-binding fragment thereof of claim 1, wherein the VH comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 34, 35, and 36, respectively, and the VL comprises CDRs 1, 2, 3 with the amino acid sequences set forth in SEQ ID NOs: 31, 32, and 33, respectively.

6. The antibody or antigen-binding fragment thereof of claim 1, wherein the antibody or antigen-binding fragment specifically binds to human PD-1.

7. The antibody or antigen-binding fragment thereof of claim 1, wherein the antibody or antigen-binding fragment is a humanized antibody or antigen-binding fragment thereof.

8. The antibody or antigen-binding fragment thereof of claim 1, wherein the antibody or antigen-binding fragment is a single-chain variable fragment (scFv) or a multi-specific antibody.

9. A nucleic acid comprising a polynucleotide encoding a polypeptide comprising: (1) an immunoglobulin heavy chain or a fragment thereof comprising a heavy chain variable region (VH) comprising complementarity determining regions (CDRs) 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 8, 9, and 10, respectively, and wherein the VH, when paired with a light chain variable region (VL) comprising the amino acid sequence set forth in SEQ ID NO: 2 binds to PD-1; (2) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 5, 6, and 7, respectively, and wherein the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 4 binds to PD-1; (3) an immunoglobulin heavy chain or a fragment thereof comprising a heavy chain variable region (VH) comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 22, 23, and 24, respectively, and wherein the VH, when paired with a light chain variable region (VL) comprising the amino acid sequence set forth in SEQ ID NO: 16 binds to PD-1; (4) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 19, 20, and 21, respectively, and wherein the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 18 binds to PD-1; (5) an immunoglobulin heavy chain or a fragment thereof comprising a heavy chain variable region (VH) comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 74, 75, and 76, respectively, and wherein the VH, when paired with a light chain variable region (VL) comprising the amino acid sequence set forth in SEQ ID NO: 68 binds to PD-1; (6) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 71, 72, and 73, respectively, and wherein the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 70 binds to PD-1; (7) an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 34, 35, and 36, respectively, and wherein the VH, when paired with a VL comprising the amino acid sequence set forth in SEQ ID NO: 28 binds to PD-1; (8) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 31, 32, and 33, respectively, and wherein the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 30 binds to PD-1; (9) an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 54, 55, and 56, respectively, and wherein the VH, when paired with a VL comprising the amino acid sequence set forth in SEQ ID NO: 48 binds to PD-1; (10) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 51,

52, and 53, respectively, and wherein the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 50 binds to PD-1; (11) an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 64, 65, and 66, respectively, and wherein the VH, when paired with a VL comprising the amino acid sequence set forth in SEQ ID NO: 58 binds to PD-1; (12) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 61, 62, and 63, respectively, and wherein the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 60 binds to PD-1; (13) an immunoglobulin heavy chain or a fragment thereof comprising a VH comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 44, 45, and 46, respectively, and wherein the VH, when paired with a VL comprising the amino acid sequence set forth in SEQ ID NO: 38 binds to PD-1; or (14) an immunoglobulin light chain or a fragment thereof comprising a VL comprising CDRs 1, 2, and 3 comprising the amino acid sequences set forth in SEQ ID NOs: 41, 42, and 43, respectively, and wherein the VL, when paired with a VH comprising the amino acid sequence set forth in SEQ ID NO: 40 binds to PD-1.

10. A vector comprising one or more of the nucleic acids of claim 9.

11. A cell comprising one or more of the nucleic acids of claim 9.

12. A method of producing an antibody or an antigen-binding fragment thereof, the method comprising culturing the cell of claim 11 under conditions sufficient for the cell to produce the antibody or the antigen-binding fragment.

13. An antibody or antigen-binding fragment thereof that binds to PD-1 comprising a heavy chain variable region (VH) comprising SEQ ID NO: 4, 18, 30, 40, 50, 60, or 70, and a light chain variable region (VL) comprising SEQ ID NO: 2, 16, 28, 38, 48, 58 or 68.

14. The antibody or antigen-binding fragment thereof of claim 13, wherein the VH comprises the sequence of SEQ ID NO: 4 and the VL comprises the sequence of SEQ ID NO: 2.

15. The antibody or antigen-binding fragment thereof of claim 13, wherein the VH comprises the sequence of SEQ ID NO: 18 and the VL comprises the sequence of SEQ ID NO: 16.

16. The antibody or antigen-binding fragment thereof of claim 13, wherein the VH comprises the sequence of SEQ ID NO: 70 and the VL comprises the sequence of SEQ ID NO: 68.

17. The antibody or antigen-binding fragment thereof of claim 13, wherein the VH comprises the sequence of SEQ ID NO: 30 and the VL comprises the sequence of SEQ ID NO: 28.

18. The antibody or antigen-binding fragment thereof of claim 13, wherein the VH comprises the sequence of SEQ ID NO: 50 and the VL comprises the sequence of SEQ ID NO: 48.

19. The antibody or antigen-binding fragment thereof of claim 13, wherein the VH comprises the sequence of SEQ ID NO: 60 and the VL comprises the sequence of SEQ ID NO: 58.

20. The antibody or antigen-binding fragment thereof of claim 13, wherein the VH comprises the sequence of SEQ ID NO: 40 and the VL comprises the sequence of SEQ ID NO: 38.

21. An antibody or antigen-binding fragment thereof comprising VH CDRs 1, 2, 3, and VL CDRs 1, 2, 3 of an antibody or antigen-binding fragment thereof comprising a heavy chain variable region (VH) comprising SEQ ID NO: 4, 18, 30, 40, 50, 60, or 70, and a light chain variable region (VL) comprising SEQ ID NO: 2, 16, 28, 38, 48, 58 or 68.

22. An antibody-drug conjugate comprising the antibody or antigen-binding fragment thereof of claim 1 covalently bound to a therapeutic agent.

23. A method of treating a subject having cancer, the method comprising administering a therapeutically effective amount of a composition comprising the antibody or antigen-binding fragment thereof of claim 1, to the subject.

24. The method of claim 23, wherein the subject has a solid tumor, a hematologic cancer, lymphoma, melanoma, non-small cell lung cancer, head and neck squamous cell carcinoma, relapsed or refractory classical Hodgkin lymphoma, squamous cell lung cancer, renal cell carcinoma, cutaneous squamous cell carcinoma, urothelial carcinoma, merkel-cell carcinoma, or

mesothelioma.
25. A method of decreasing the rate of tumor growth, the method comprising contacting an immune cell with an effective amount of a composition comprising an antibody or antigen-binding fragment thereof of claim 1, to the subject.
26. A method of killing a tumor cell, the method comprising contacting an immune cell with an effective amount of a composition comprising the antibody or antigen-binding fragment thereof of claim 1, to the subject.
27. A method of treating or reducing the risk of developing an infectious disease, the method comprising administering a therapeutically effective amount of a composition comprising the antibody or antigen-binding fragment thereof of claim 1, to the subject.
28. A pharmaceutical composition comprising the antibody or antigen-binding fragment thereof of claim 1, and a pharmaceutically acceptable carrier.